

Empirical conversion between teleseismic magnitudes (m_b and M_s) and moment magnitude (M_w) at the Global, Euro-Mediterranean and Italian scale

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Accepted 2014 July 7. Received 2014 July 7; in original form 2013 October 21

SUMMARY

We analysed the conversion problem between teleseismic magnitudes (M_s and m_b) provided by the Seismological Bulletin of the International Seismological Centre and moment magnitudes (M_w) provided by online moment tensor (MT) catalogues using the chi-square general orthogonal regression method (CSQ) that, differently from the ordinary least-square regression method (OLS), accounts for the measurement errors of both the predictor and response variables. To account for the non-linearity of the relationships, we used two types of curvilinear models: (i) the exponential model (EXP), recently proposed by the authors of the Global Catalogue sponsored by the Global Earthquake Model (GEM) Foundation and (ii) a connected bilinear (CBL) model, similar to that proposed by Ekström & Dziewonski, where two different linear trends at low and high magnitudes are connected by an arc of circle that preserves the continuity of the function and of its first derivative at the connecting points. For M_s , we found that the regression curves computed for a global data set (GBL) are likely to be biased by the incompleteness of global MT catalogues for $M_w < 5.0$ – 5.5 . In fact, the GBL curves deviate significantly from a similar regression curve computed for a Euro-Mediterranean data set (MED) integrated with the data provided by two regional MT catalogues including many more events with $M_w < 5.0$ – 5.5 . The GBL regression curves overestimate the M_w proxies computed from M_s up to 0.5 magnitude units. Hence for computing M_w proxies at the global scale of $M_s \leq 5.5$, we suggest to adopt the coefficients obtained from the MED regression. The analysis of the frequency–magnitude relationship of the resulting M_w proxy catalogues confirms the validity of this choice as the behaviour of b -value as a function of cut-off magnitude of the GBL data set is much more stable using such approach. The incompleteness of M_w 's provided from MT global catalogues also affects the m_b GBL data set but in this case the use of the CSQ regression method, in place of the OLS, mitigates the bias and then, at low magnitudes, the EXP regression curve computed from the more complete MED data set almost coincides with that computed from the GBL data set. Our results also indicate that the slope at low magnitudes of the M_w – M_s relationship is substantially consistent with the hypothesized theoretical value of $2/3$ for $M_s < 5.0$ while the slope of the M_w – m_b relationship at high magnitudes probably reaches the theoretically expected value of 2 only in the proximity of the upper limit of m_b determinations in our data set ($m_b = 7.2$).

Key words: Earthquake source observations; Statistical seismology.

1 INTRODUCTION

Several authors have addressed the problem of converting teleseismic magnitudes (M_s and m_b) to moment magnitude (M_w) by calculating empirical relationships (mostly linear) through the ordinary least-squares (OLS) regression method (e.g. Heaton *et al.* 1986; Johnston 1996; Scordilis 2006) as well as through general orthogonal regression (GOR) methods that are more appropriate than OLS since the uncertainty of teleseismic magnitudes is not negligible when compared to that of M_w . In particular, Castellaro *et al.* (2006) have shown that the use of the GOR method (Fuller 1987) in magnitude conversions prevents a bias of the frequency–magnitude distribution law (Gutenberg & Richter 1944) in the resulting converted catalogue that instead occurs if the OLS method is used.

The GOR method by Fuller (1987) has been applied to magnitude conversions both at the global and regional scale (Bormann *et al.* 2007; Ristau 2009; Wang *et al.* 2009; Deniz & Yucemen 2010; Das *et al.* 2011, 2012; Baruah *et al.* 2012; Gasperini *et al.* 2012). Another general regression method, based on the minimization of the chi-square (CSQ) statistics, was previously proposed by Stromeyer *et al.* (2004) and later used by Grünthal *et al.* (2009), Grünthal & Wahlström (2012) and Gasperini *et al.* (2013a,b). A third method, the total weighed least squares (Krystek & Anton 2007), was used instead by Bethman *et al.* (2011) for magnitude conversions in Switzerland. Although the formulations of the three methods may appear different, Lolli & Gasperini (2012) demonstrated that, in the common case in which the ratio between the variances of the dependent and independent variables η is constant, they all provide virtually identical regression coefficients and very similar uncertainties.

Recently, Wason *et al.* (2012) proposed a procedure to estimate the true values of the independent variable to be used for computing the M_w proxies from the regression coefficients computed by the GOR method. As argued by Gasperini & Lolli (2013), such modified orthogonal regression (MGOR) method is wrong for a number of theoretical and empirical reasons. In particular, the supposedly better fit reported by Wason *et al.* (2012) between observed and computed M_w is due to their use, as goodness-of-fit criterion, of the simple standard deviation that, by definition, neglects the presence of errors of the independent variable.

The problem of converting teleseismic magnitudes to M_w has been recently addressed by Zúñiga & Figueroa-Soto (2012) using a completely different approach that assumes the temporal stability of a - and b -values of the Gutenberg & Richter (1944) law of the magnitude distribution. Such methodology, which is particularly suited for cases when estimates of quality paired magnitudes are unavailable, can provide complementary information to the more straightforward method adopted in this work.

One of the most critical aspects in the application of orthogonal regression methods to magnitude conversions concerns the estimation of the variances of the two variables. In fact, the uncertainties of magnitude determinations are not provided usually by online databases and seismic catalogues. Most papers concerning the conversion between magnitudes by GOR methods assume that the respective uncertainties are approximately equal to each other and therefore the variance ratio is $\eta = 1$. In this case, Castellaro & Bormann (2007) showed that the orthogonal regression is preferable than the ordinary one anyhow. However, the assumption of equal variances is not correct in case of regressions between M_w and traditional magnitudes as the uncertainty of M_w is of the order of 0.05–0.15 magnitude units (m.u.; Helffrich 1997; Kagan 2002, 2003; Gasperini *et al.* 2012) whereas the uncertainty of traditional magnitudes is of the order of 0.2–0.3 m.u. (Bormann *et al.* 2013; Gasperini *et al.* 2013b).

Gasperini *et al.* (2013a) approached the problem to estimate the uncertainties associated with various types of local magnitude determined in Italy from 1981 to 2010 by first determining initial values of the variances using available *a priori* information for the various data sets and then adjusting them iteratively so as to make the standard deviation resulting from *a priori* variances to coincide with the empirical standard deviation of regression residuals. The validity of this approach has been confirmed by the fact that the correction to apply to the initial value was almost negligible, for the unique data set (Margottini *et al.* 1993) for which a reliable *a priori* estimate of local magnitude (M_L) uncertainty was available. In this paper, we proceeded similarly, basing on the standard deviations deduced from the distribution of magnitude measures and also considering further sources of errors related to heterogeneities of wave propagation properties of the Earth's crust and lithosphere in different regions.

2 FORM OF THE RELATIONSHIP BETWEEN M_w AND TELESEISMIC MAGNITUDES

It is well known that the relationship between moment magnitude M_w and surface magnitude M_s is not linear. From theoretical considerations on the excitation of surface waves with period of about 20 s, from which M_s magnitudes are computed, the scaling coefficient between the decimal logarithm of scalar seismic moment M_0 and M_s should be about 1 for small earthquakes, about 3/2 for intermediate magnitude ones and about 3 for the biggest ones (Kanamori & Anderson 1975). Basing on the IASPEI (2005) formula for M_w

$$M_w = \frac{2}{3} (\log_{10} M_0 - 9.1) \quad (1)$$

the slopes of the linear relation between M_w and M_s should be 2/3, 1 and 2, respectively. It is usually assumed that the changing point between slopes 2/3 and 1 occurs within the range $6 < M_s < 7$ (e.g. Ekström & Dziewonski 1988; Scordilis 2006; Das *et al.* 2011; Storchak *et al.* 2013) whereas the transition magnitude between slopes 1 and 2 is not well known because of the scarceness of very large earthquakes, but it should occur for $M_s > 8$ (e.g. Scordilis 2006; Das *et al.* 2011; Storchak *et al.* 2013). To model the non-linearity of the relationship between M_w and M_s , Ekström & Dziewonski (1988) proposed a square root function, within the range $5.3 \leq M_s \leq 6.8$, which connects two straight lines with slope 2/3 and 1 outside such range.

It is also well known that the band-limited short-period body wave magnitude m_b saturates at values larger than about 5.5–6.0. According to Kanamori & Anderson (1975) the linearity of the relationship is also arguable at very low magnitudes. Hence a linear relation for modelling the relationship between M_w and m_b over the entire range of magnitudes is certainly inappropriate.

Even for the relationship between M_w and m_b we should recognize three traits with slopes of 2/3, 1 and 2 but, owing to the lower period of waves considered for estimating m_b (about 1 s), the change points should occur at lower values than for M_s (Kanamori & Anderson 1975). In particular, the observed saturation for $m_b > 5.5$ –6.0 should correspond to the transition from slope 1 to 2. For lower m_b the relationship should be about 1:1 (Bormann *et al.* 2009) down to a lower threshold that is presently not known but we might argue it should be lower than the minimum of current m_b determinations (about $m_b = 3.0$ –3.5).

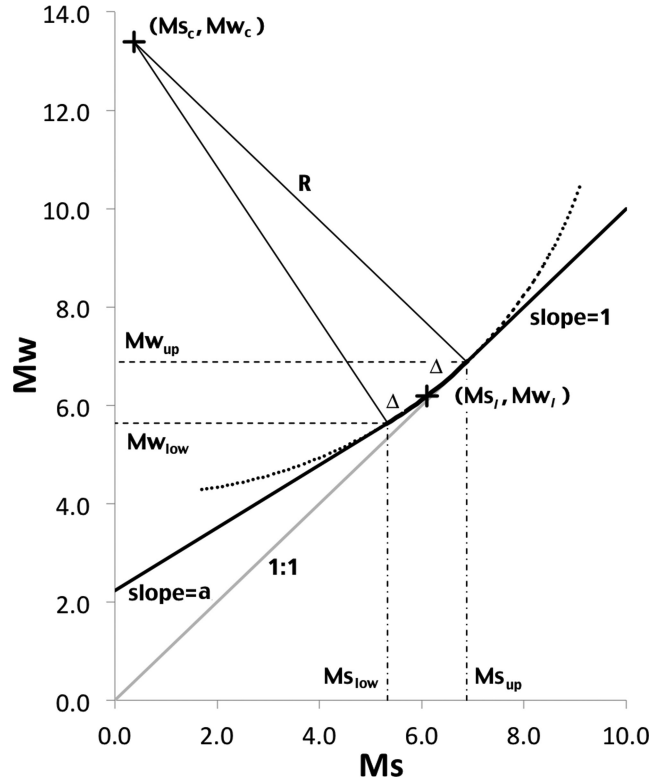


Figure 1. Geometrical construction of the CBL model (see text and Appendix B for the meaning of symbols).

Very recently, in the framework of the initiative promoted by the Global Earthquake Model (GEM) Foundation for the revision and homogenization of the Global Instrumental Earthquake Catalogue (1900–2009), Storchak *et al.* (2013) proposed an exponential regression model for converting to M_w both M_s and m_b provided by a revised version of the Seismological Bulletin of the International Seismological Centre (ISC)

$$M_w = \exp(a + bx) + c, \quad (2)$$

where a , b and c are empirical parameters determined by regressions of available M_w observations for which an M_s or an m_b (x) is also available. The two advantages of such approach are that (i) a unique regression, using only three free parameters, can model the non-linearity of the relationships between M_w and teleseismic magnitudes and (ii) the exponential fit appears to work better than a linear GOR for earthquakes with $M_w < 7$. Unfortunately, Storchak *et al.* (2013) applied their new regression model using an ordinary least-square approach (minimizing the simple standard deviation between observed and predicted M_w) that, as noted before, is not appropriate for regressions between M_w and teleseismic magnitudes.

Gasparini *et al.* (2013b) have shown how such exponential model can be applied using the CSQ method that minimize the chi-square, that is the ratio between the observed and the *a priori* variances of predicted M_w (see Appendix A). In Appendix A, we also describe how consistent prediction errors can be estimated for each M_w proxy computed by EXP regression coefficients.

Storchak *et al.* (2013) did not give any physical explanation for their exponential model, hence its adoption was purely empirical. Moreover, such model has unrealistic implications as for example that the slope tends to 0 (horizontal) for very small magnitudes and to ∞ (vertical) for very large magnitudes whereas, according to the theory, the slope should range from 2/3 to 2.

Another possible functional model for M_s , coherent with the physical description by Kanamori & Anderson (1975), is that proposed by Ekström & Dziewonski (1988). In this work, we preferred a slightly different formulation that has a simple geometrical description but is very close to that proposed by Ekström & Dziewonski (1988). In our regression model (see Fig. 1), M_s coincides with M_w above a certain upper magnitude threshold $M_{s\text{ up}}$, and is linearly related to M_w (with slope $a < 1$) below a certain lower magnitude threshold $M_{s\text{ low}}$. In the interval between $M_{s\text{ low}}$ and $M_{s\text{ up}}$ the two straight lines are connected by an arc of circle that preserves the continuity of the function and of its first derivative at the changing points

$$M_w(M_s) = \begin{cases} aM_s + b, & M_s < M_{s\text{ low}} \\ M_{wc} - \sqrt{R^2 - (M_s - M_{sc})^2}, & M_{s\text{ low}} \leq M_s \leq M_{s\text{ up}} \\ M_s, & M_s > M_{s\text{ up}} \end{cases} \quad (3)$$

where a and b are, respectively the slope and the intercept of the straight line at low magnitudes, while M_{sc} , M_{wc} and R are, respectively, the coordinates of the centre and the radius of the connecting circular arc in the M_s – M_w Cartesian space (Fig. 1). The connecting points between the straight lines and the arc are located symmetrically from the intersection point at a distance Δ along the straight lines. Owing to the assumptions of continuity at the changing points, M_{sc} , M_{wc} and R can be computed as a function of a , b and Δ (see Appendix B). Hence, the fitting of our circularly connected bilinear (CBL) model requires the numerical estimation of three parameters. In Appendix B, we also show how such model can be regressed using the CSQ method and even how consistent error estimates of M_w proxies determined using the regressed coefficients could be computed.

Note that the arbitrariness of the choice of a circular function to connecting the two rectilinear traits is not very relevant in the practice as the subtended arcs will range from 10° to 15° . Hence we confidently expect that other possible functional forms may not deviate considerably from the circle over so small arcs.

The closeness to the 1:1 line of the m_b – M_w relationship at magnitudes lower than about 5.0–5.5 (Bormann *et al.* 2009; Gasperini *et al.* 2013b) suggests the introduction of a reversed circularly connected bilinear model (CBL-R) for m_b where M_w is assumed to coincide with m_b below a certain lower magnitude threshold $m_{b\text{ low}}$ and is linearly related to M_w (with slope $a > 1$) above an upper magnitude threshold $m_{b\text{ up}}$. As for the previous CBL model, the two traits are connected by an arc of circle that preserves the continuity of the function and of first derivative at the change points

$$\begin{aligned} M_w &= m_b & m_b < m_{b\text{ low}} \\ M_w &= M_{wc} - \sqrt{R^2 - (m_b - m_{bc})^2} & m_{b\text{ low}} \leq m_b \leq m_{b\text{ up}}, \\ M_w &= am_b + b & m_b > m_{b\text{ up}} \end{aligned} \quad (4)$$

where again a and b are, respectively, the slope and the intercept of the straight line at high magnitudes, while m_{bc} , M_{wc} and R are, respectively, the coordinates of the centre and the radius of the connecting arc in the m_b – M_w Cartesian space. Even in this case the connecting points between the straight lines and the arc are located symmetrically from the intersection point at a distance Δ along the straight lines. Some details on the implementation using the CSQ regression method are described in Appendix C.

3 BUILDING OF THE MAGNITUDE DATA SETS TO REGRESS

The ISC online Bulletin (available at <http://www.isc.ac.uk/iscbulletin/search/bulletin/>, last accessed August 2014) is the most complete source of earthquake locations and magnitudes at the global scale. It reports both definitive (revised) and preliminary (unchecked) locations by a merged data set of arrival times provided by a number of global, regional, and local contributing observatories (virtually all of the seismological institutions of the World). It also reports M_s and m_b computed according to international standards (IASPEI 2005; Bormann *et al.* 2013) using the amplitude and period data provided by contributor institutions. ISC actually provides even the locations and magnitudes computed by contributor institutions but such data are heterogeneous and basically unreliable. Hence we did not consider such contributed magnitudes in our analyses. The time window of the definitive ISC Bulletin presently ranges from 1964 to the beginning of 2011 but for the comparison of teleseismic magnitudes with M_w , the effective starting date coincides with the first M_w determinations available from the Global Centroid Moment Tensor catalogue in 1976. For reasons of database stability during our analysis we chose year 2010 as ending date. For our calibrations we considered unreliable and then discarded all M_s and m_b estimates made using less than three stations.

We constructed an integrated M_w (IMW) reference data set combining M_w estimates from the moment tensor catalogues available globally and for the European Mediterranean and Italian areas:

- (1) The Global Centroid Moment Tensor (GCMT) catalogue from 1976 to 2010 (Dziewonski *et al.* 1981; Ekström *et al.* 2012) available at <http://www.globalcmt.org/CMTfiles.html> (last accessed August 2014).
- (2) The USGS-NEIC MT (NEIC) catalogue from 1980 to 2009 (Sipkin 1982, 1994), available at <http://earthquake.usgs.gov/earthquakes/eqarchives/sopar/> (last accessed December 2012).
- (3) The European-Mediterranean Regional Centroid Moment Tensor (RCMT) catalogue of the Istituto Nazionale di Geofisica e Vulcanologia (INGV) of Rome from 1997 to 2010 (Pondrelli *et al.* 2002, 2011), available at <http://www.bo.ingv.it/RCMT/> (last accessed August 2014).
- (4) the Eidgenössische Technische Hochschule Zürich (ETHZ) Regional MT catalogue from 1999 to 2006 (Braunmiller *et al.* 2002), available at http://www.seismo.ethz.ch/prod/tensors/mt_oldcat/index_EN (last accessed August 2014).
- (5) The Time Domain Moment Tensor (TDMT) catalogue of INGV from 2006 to 2010 (Scognamiglio *et al.* 2009), available at <http://cnt.rm.ingv.it/tdmt.html> (last accessed August 2014).

The details of the IMW catalogue construction are given in Gasperini *et al.* (2012). In this work, for the GBL area (extended to the entire World), we only considered the GCMT and NEIC catalogues. For the MED area (Latitude 25°N – 60°N , Longitude 20°W – 50°E) we added the data from the RCMT and ETHZ catalogues. For the ITA area (Latitude 35°E – 48°E , Longitude 6°N – 20°N) we also added the data from the TDMT catalogue. Note that the MED area coincides with that adopted by Gasperini *et al.* (2012) but it is somehow smaller than that used by Gasperini *et al.* (2013b).

According to Gasperini *et al.* (2012) we assumed as reference the M_w estimates from GCMT and RCMT catalogues (which were found to be equivalent to each other) and corrected, by uniform shifts, the M_w estimates from NEIC (+0.05 m.u.), ETHZ (−0.05 m.u.) and TDMT (+0.20 m.u.) catalogues. M_w estimates from GCMT and NEIC were discarded when other catalogues report for the same earthquake $M_w < 5.4$ or $M_w > 7.0$, respectively (see Gasperini *et al.* (2012) for details). In cases when M_w estimates from more than one catalogue were available for the same earthquake, we assumed as best M_w estimate their weighted mean, using as natural weight the inverse of the respective variances.

We matched the ISC records with those of the IMW data set based on the origin time difference (within 10 s) and the epicentral distance (within 20 km). These thresholds were chosen small enough to avoid magnitudes of different earthquakes being matched together. Therefore, it is possible that some M_w – m_b pairs were missed.

4 ESTIMATION OF INITIAL UNCERTAINTIES OF M_s AND m_b

The ISC Bulletin provides M_s and m_b ‘errors’ only for earthquakes occurred since 2009 when the procedure was changed by introducing robust estimators (median) for computing average magnitudes. The error e_i reported for each i th earthquake is computed as the standard median absolute deviation (SMAD) from the median of the individual estimates made by different stations (Bondar & Storchak 2011). Assuming a Normal distribution of data, the standard deviation σ can be computed from the SMAD as (Russeeuw & Croux 1993)

$$\sigma_i = 1.4826e_i. \quad (5)$$

Actually, such standard deviation represents the dispersion of the sample of individual magnitude estimates for a given earthquake rather than the uncertainty of the mean magnitude. As the ISC Bulletin also provides the numbers of stations n_i used to compute the magnitude, the standard uncertainty of the mean could be computed as

$$\bar{\sigma}_i = \frac{\sigma_i}{\sqrt{n_i}}. \quad (6)$$

Unfortunately, this uncertainty could be estimated only for earthquakes occurred in 2009 and 2010 but not for those occurred before. We then proceeded similarly to Gasperini *et al.* (2013a) by computing average standard deviations for the whole set of N earthquakes from 2009 to 2010 (with errors reported). We summed all of the squared standard deviations multiplied by the numbers of observations from different stations and then took the square root of such sum divided by the total number of magnitude observations

$$\sigma = \sqrt{\frac{\sum_{i=1}^N \sigma_i^2 n_i}{\sum_{i=1}^N n_i}}. \quad (7)$$

These computations give average standard deviations of 0.33 m.u. for M_s and of 0.36 m.u. for m_b . We also performed the same analysis on a set of standard deviations computed from data of individual stations that we downloaded from the ISC site for sampling periods of few months in some years between 1970 and 2008. Such analysis exactly confirmed the average standard deviation of 0.33 m.u. for M_s and gave a slightly larger estimate of 0.41 m.u. for m_b . As we can expect that the quality of magnitude determinations improved with time, we assumed the latter largest value as representative of the dispersion of m_b measures from the ISC Bulletin over the entire time window we consider.

From such estimates, the standard error of the mean magnitude could be computed by applying eq. (6). As the strongest earthquakes are observed by hundreds of stations all over the World this brings in some cases to estimating uncertainties of the order of a few hundreds of m.u. that are clearly unrealistic because the observed residuals can even be a factor of 5–10 larger.

Actually, such uncertainties only describe the random variability of magnitude measures for each given earthquake but do not include other sources of error that may affect magnitude estimates. Teleseismic magnitudes are indirect measures computed using attenuation equations with epicentral distance given by the *Prague formula* (Vanek *et al.* 1962) for M_s and by the tables and nomograms reported by Gutenberg & Richter (1956) for m_b . Such calibration equations and tables were established, at the time, from the analysis of global data sets and thus are representative of the average attenuation properties of the Earth. It is well known, however, that the propagation properties of relatively short-period seismic waves (≈ 20 s for M_s and ≈ 1 s for m_b) may vary significantly in regions characterized by different crustal and lithospheric thickness and heat flow.

Note that this problem does not affect significantly moment magnitudes M_w from global data sets (GCMT and NEIC) because they are determined using very long period waves (up to 350 s) that are scarcely influenced by the local properties of the crust and the lithosphere. On the other hand, moment magnitudes M_w from regional and national data set (RCMT, ETHZ and TDMT), which are determined using shorter period waves, do not show significant scale deviations and can be made consistent with global data sets by simple offset corrections (Gasperini *et al.* 2012). The problem does not affect either local magnitude M_L provided that the attenuation table used to determine it is properly calibrated, according to Richter (1935), in the area where M_L is computed.

To evaluate the variability of magnitude estimates due to the regional deviations from the global average propagation properties, we computed the average residuals between M_w and the proxies of M_s and m_b (computed using the EXP coefficients from a preliminary global

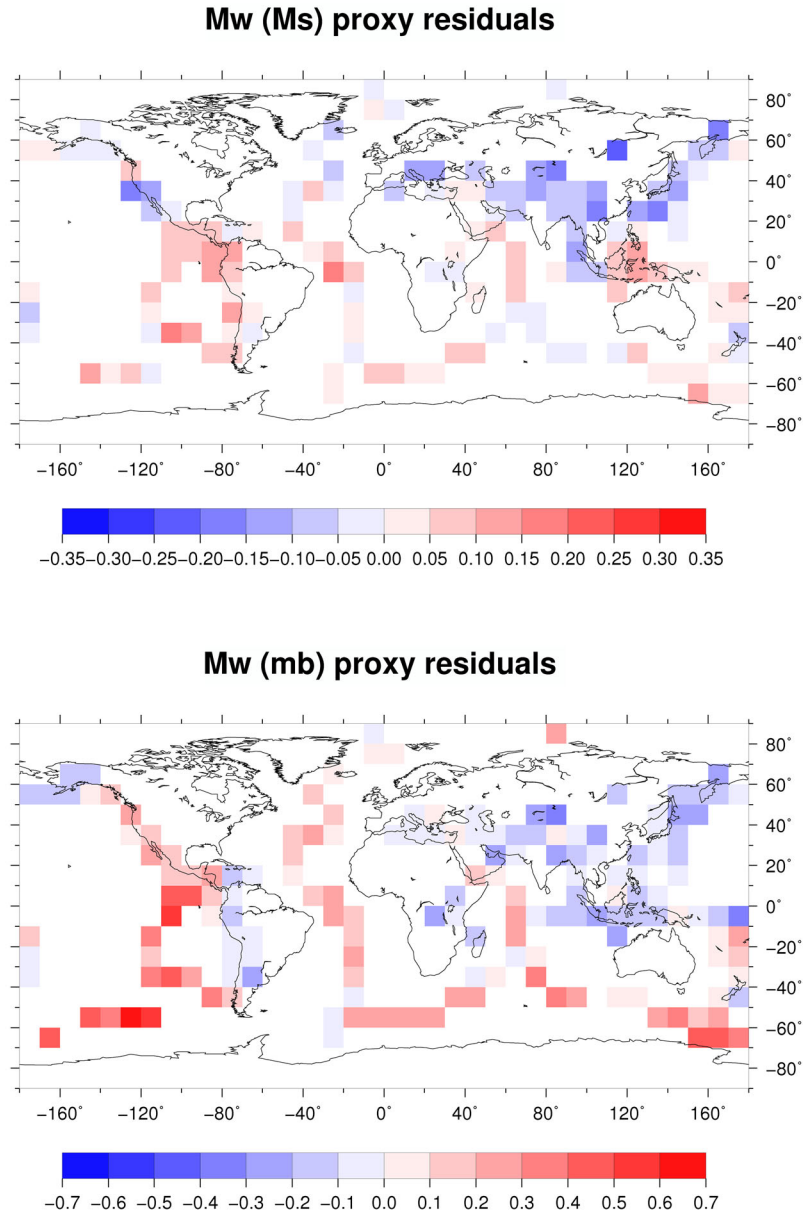


Figure 2. Spatial distribution of average M_w proxy residuals for M_s (top panel) and m_b (bottom panel) over cells of 10° . Positive residuals (red) indicate that observed M_w 's are larger than proxies. Only the cells with more than 20 earthquakes are plotted.

regressions) over a tessellation of the globe with cells sizes varying from 5° to 30° . In Figs 2(a) and (b) we show the map of such average residuals for cell sizes of 10° indicating, for both M_s and m_b , a prevalence of negative residuals (teleseismic proxies overestimate observed M_w) in continental shields and subduction zones and of positive residuals (teleseismic proxies underestimate observed M_w) in mid-oceanic ridges. This confirms our hypothesis that magnitude estimates are influenced by the regional propagation properties (less attenuating in the continental shields and more attenuating in the ocean ridges). Even the large extension of the areas where the average residuals have agreeing signs indicates that the deviations from the average are not random and are likely to be related to physical reasons.

In Table 1 we report the global standard deviations of M_w average residuals in cells with more than 20 earthquakes for different tessellation sizes. The stability around values of $\sigma_{M_s G} = 0.08$ m.u. for M_s and $\sigma_{m_b G} = 0.20$ m.u. for m_b indicates that they likely reflect the real variability of M_s and m_b at the global scale and thus can be used to estimate the uncertainty of magnitudes due to regional variations of the attenuation properties. Moreover, as the heterogeneities of the propagation properties may occur at scales smaller than 5° , we can expect that such standard deviations represent a lower bound of the real dispersion of average residuals.

As standard deviations are computed using 20 earthquakes at least, we can assume that they are scarcely affected by the random scatter of magnitude estimated from different stations. Hence, the statistical uncertainties described by eq. (6) (using the standard deviations of 0.33 and 0.41 m. u. previously estimated for M_s and m_b measures, respectively) have to be added to the standard deviations related to regional

Table 1. Standard deviations of residuals between observed and proxy M_w at the GBL scale, for M_s ($\sigma_{m_{sG}}$) and m_b ($\sigma_{m_{bG}}$) as a function of the cell size of geographical tessellation.

Cell size	$\sigma_{m_{sG}}$	$\sigma_{m_{bG}}$
5°	0.084	0.206
10°	0.077	0.207
15°	0.073	0.202
20°	0.070	0.201
30°	0.060	0.187

attenuation heterogeneities using the error propagation law. In summary, the initial uncertainty of each magnitude determination is established as follows:

$$\sigma_{M_s} = \sqrt{\frac{0.33^2}{n} + \sigma_{m_{sG}}^2}$$

$$\sigma_{m_b} = \sqrt{\frac{0.41^2}{n} + \sigma_{m_{bG}}^2}, \quad (8)$$

where n is the number of stations used for each magnitude determination.

5 ANALYSIS OF THE M_w – M_s RELATIONSHIP

For the GBL data set we have 19 475 earthquakes in the range $3.3 \leq M_s \leq 8.7$ with both M_s and M_w . In Fig. 3, we indicate with red and black lines the CSQ regression curves computed for the EXP and CBL models, respectively, after adjustment of $\sigma_{m_{sG}}$ so that the *a priori* and the empirical regression standard deviations coincide (see Table 2). The diagonal grey line indicates the equality between M_w and M_s and the two vertical grey lines correspond to the transition magnitudes ($M_{s \text{ low}}$ and $M_{s \text{ up}}$) between the two straight lines and the connecting arc of the CBL model.

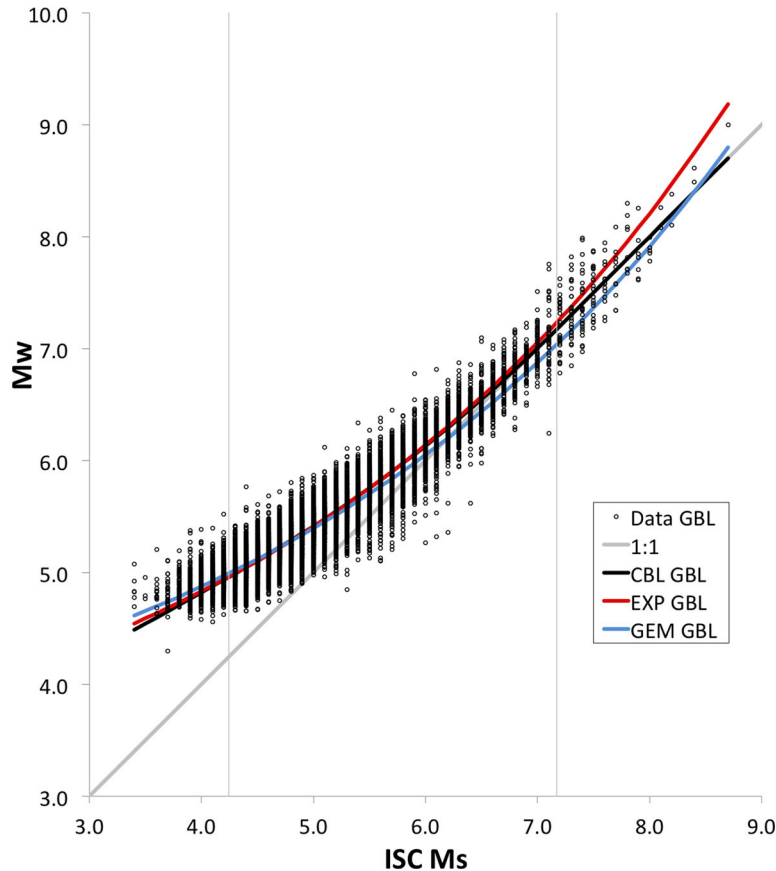


Figure 3. M_w – M_s data distribution and regression curves for the GBL data set. The two vertical grey lines indicate the transition magnitudes ($M_{s \text{ low}}$ and $M_{s \text{ up}}$) between the linear traits and the connecting arc of the CBL model.

Table 2. Initial and resized standard deviations of M_s ($\sigma_{M_s G}$) and m_b ($\sigma_{m_b G}$) average residuals due to spatial variability of attenuation equations for various data sets.

	Initial	Resized GBL	Resized MED	Resized ITA
$\sigma_{M_s G}$	0.08	0.14	0.16	0.15
$\sigma_{m_b G}$	0.20	0.23	0.17	0.14

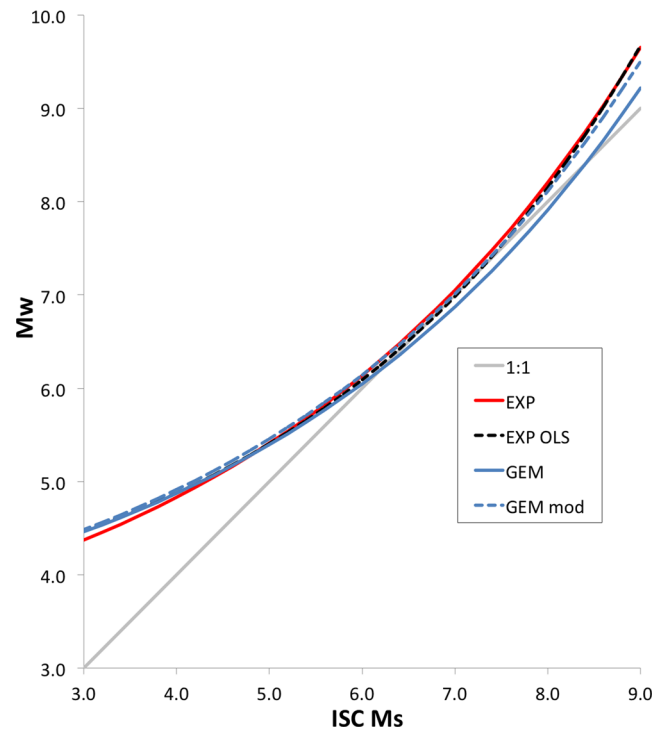


Figure 4. Comparison between different EXP M_w – M_s regression curves for the GBL data set. GEM mod (dashed blue line) indicates the curve computed using a coefficient b slightly modified (0.2349) with respect to the value (0.23) reported by Storchak *et al.* (2013).

The two models almost coincide with each other over the magnitude range $3.5 < M_s < 7.0$ and deviate from one another at larger and lower magnitudes. The slope of the CBL model at low magnitudes (0.560 ± 0.005) is definitely lower than the theoretical value of $2/3$ suggested by Kanamori & Anderson (1975).

The progressive deviation of the EXP curve (red) from the 1:1 line for $M_s > 7.5$ might indicate the onset of M_s saturation and of the transition to the trait with slope 2 hypothesized by Kanamori & Anderson (1975). In Fig. 3 we can note however that the EXP curve overestimates the M_w of all of the earthquakes with $M_s > 8.0$. Hence we might argue that, to some extent, such deviation is an effect of the distribution of data at lower magnitudes and on the numerical properties of the purely empirical EXP model.

We also plotted in Fig. 3 the curve (solid blue) representing the EXP model computed by Storchak *et al.* (2013), for the GEM global catalogue (GEM GBL), using the OLS regression method. We can note how the latter coincides with the other two curves only within a narrower band of magnitudes ($4.5 < M_s < 5.5$) and deviates from them, and particularly from our EXP model up to about 0.3–0.4 m.u., at large magnitudes.

We might argue that the deviation of our EXP model with respect to that computed by Storchak *et al.* (2013) might be due to the different magnitude data set used by them (slightly modified with respect to the ISC database available online) and only to a minor extent to the use of the OLS regression method because the OLS regression computed using our data sets (dashed black in Fig. 4) does not substantially deviates from the CSQ (red) at large magnitudes. However, we may observe that the GEM curve (solid blue), plotted using the coefficients reported by Storchak *et al.* (2013) ($a = -0.22$, $b = 0.23$, $c = 4.86$), lies close to the 1:1 equality line for $M_s > 8.0$ whereas, in their fig. 5.12, it seems to stay well above such line, that is very close to our OLS regression line. We might hypothesize that this is the effect of the rounding to only two decimals of coefficient b reported by Storchak *et al.* (2013). In fact, if for example we use $b = 0.2349$ (that rounds to 0.23 using two decimals) then the resulting curve (GEM mod, dashed blue in Fig. 4) became much closer to our OLS and CSQ curves.

This indicates that to avoid rounding errors it is generally advisable to report regression coefficients with a number of digits larger than that can be considered significant based on coefficient uncertainties. This because some coefficients are negatively correlated to each other and then their individual uncertainties may largely overestimate the real uncertainty of the regression curve.

For the MED area we have 1200 earthquakes with both M_s and M_w (Fig. 5) in the range $2.9 \leq M_s \leq 7.7$. The CSQ regression curves for the EXP (solid red) and CBL (black) models almost coincide with each other up about $M_s = 6.5$. Differently from the GBL data set, the EXP

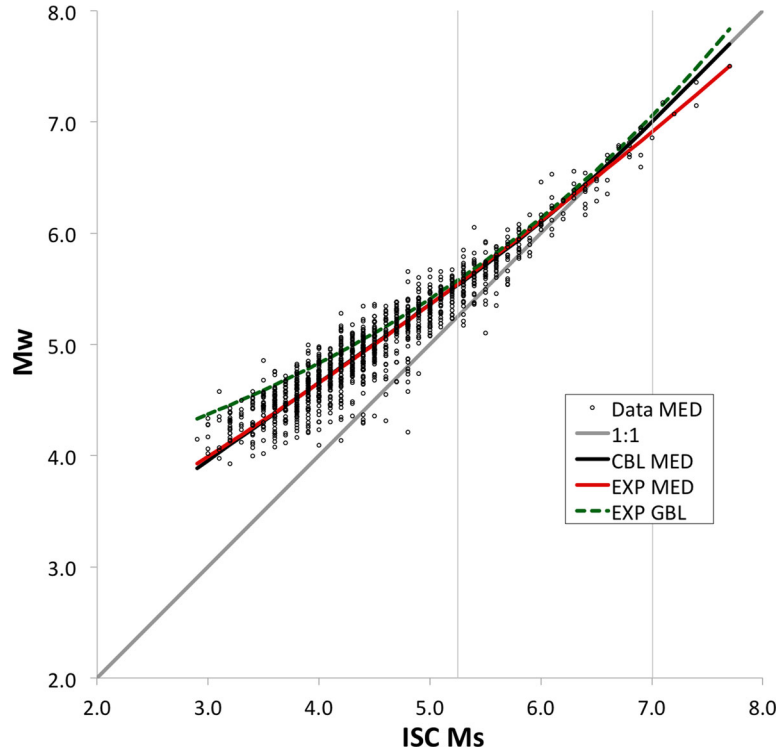


Figure 5. Same as Fig. 3 for the MED data set.

Table 3. CSQ regression parameters and standard deviations (68 per cent) of the EXP model (eq. 2) for M_s and m_b , for different data sets. σ_w indicates the weighted standard deviation of regression (see Appendix A).

Mag.	Data set	a	b	c	σ_w
M_s	GBL	-0.109 ± 0.047	0.229 ± 0.005	2.586 ± 0.065	0.1462
	MED	2.133 ± 0.343	0.063 ± 0.017	-6.205 ± 3.013	0.1703
m_b	ITA	1.421 ± 0.593	0.108 ± 0.042	-1.863 ± 2.701	0.1685
	GBL	0.741 ± 0.170	0.210 ± 0.017	-0.785 ± 0.520	0.3326
	MED	0.719 ± 0.329	0.212 ± 0.034	-0.737 ± 0.938	0.2397
	ITA	0.058 ± 0.451	0.284 ± 0.055	0.770 ± 0.770	0.2171

model lies slightly below the 1:1 line at larger magnitudes, probably because M_s is larger than M_w for all of the earthquakes with $M_s \geq 7.0$ (Fig. 5). This feature is quite surprising and might represent a significant peculiarity of the MED area but the low number of earthquakes within this magnitude range suggests some caution.

At low magnitudes both curves lies well below the EXP curve computed for the GBL set (dashed green) up to about 0.3–0.4 m. u. The slope of the CBL line at low magnitudes (0.701 ± 0.008) is slightly higher (Table 4) but substantially consistent with the theoretical value of 2/3. The deviation at low magnitudes between the MED and GBL data sets might indicate a peculiar property of the MED area with respect to the GBL one but it might even be the effect of the overestimation of magnitudes $M_w < 5.0$ –5.5 of the GCMT catalogue discussed by Gasperini *et al.* (2012) and/or of the incompleteness of the GBL data set of M_w magnitudes hypothesized by Gasperini *et al.* (2013b).

Actually, global CMT catalogues report very few earthquakes with $M_w < 5.0$ (Kagan 2002, 2003). From our regressions (Table 3), $M_w = 5.0$ corresponds to about $M_s = 4.3$, then earthquakes with M_s lower than such threshold are likely to be not included in global MT catalogues. The only earthquakes with $M_s < 4.3$ for which we can have an M_w in our GBL data set are those for which M_s is somehow underestimated. For the MED data set instead we have many more earthquakes with $M_w < 5.0$ provided by RCMT and ETHZ regional catalogues, hence such incompleteness is mitigated.

For the ITA area we have 233 earthquakes with both M_s and M_w (Fig. 6) in the range $2.9 \leq M_s \leq 6.9$. In this case, the CSQ regression curves for the EXP (solid red) and CBL (black) models almost coincide with each other over the entire magnitude range. At large magnitudes the EXP MED curve is very close to the 1:1 line and to the EXP curve computed for the GBL (dashed green) and MED (dashed red) areas. The slope of the CBL line at low magnitudes (0.678 ± 0.028) is definitely larger than that computed for GBL data sets but not very different from that computed for MED. The quite uniform negative difference of about -0.15 m. u. between ITA and MED curves in the range $3.0 < M_s < 5.5$ seems to indicate that the ITA area is slightly less attenuating than the average of the MED area.

The comparison with MED and ITA areas indicates that GBL regression curves are likely to be biased by the incompleteness of global M_w catalogues at low magnitudes ($M_w < 5.0$ –5.5) and suggests that MED and ITA regressions might better reproduce the real behaviour of

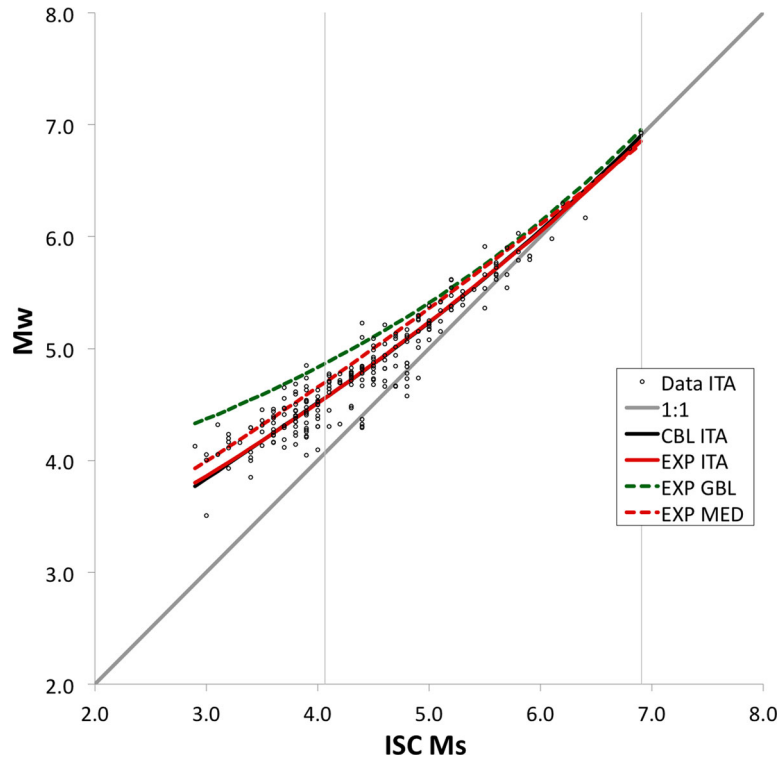


Figure 6. Same as Fig. 3 for the ITA data set.

Table 4. CSQ regression parameters, standard deviations (68 per cent) and covariances of the CBL model (eq. 3) for M_s , for different data sets. σ_w indicates the weighted standard deviation of regression (see Appendix B).

Data set	a	b	Δ	$M_{s\ c}$	$M_{w\ c}$	r	$M_{s\ low}$	$M_{s\ up}$	σ_w
GBL	0.560 ± 0.005	2.582 ± 0.022	1.854 ± 0.025	-2.291	16.636	13.383	4.24	7.17	0.1467
MED	0.701 ± 0.008	1.853 ± 0.036	1.151 ± 0.170	-2.319	16.334	13.190	5.25	7.01	0.1714
ITA	0.678 ± 0.028	1.803 ± 0.116	1.851 ± 0.262	-6.854	20.664	19.458	4.06	6.91	0.1689

the M_w – M_s relationship at low magnitudes even for the GBL catalogue. A possible option to correct the incompleteness bias of the GBL regression at low magnitudes might be using the MED coefficients even for the GBL data set for $M_s \leq 5.5$ (the difference between the two curves at the changing point $M_s = 5.5$ is almost negligible: of the order of 0.03–0.04 m.u.).

The discrepancy between GBL and MED regressions for $M_w > 7.0$ – 7.5 is not easy to interpret because of the scarce number of large earthquakes in the MED data set. We may observe that, with the aim of converting M_s to M_w , the question is not so relevant because major earthquakes are usually studied with much detail and then reliable estimates of M_w can be determined otherwise.

For all of the three areas, the EXP model fits the data slightly better than the CBL, as indicated by the lower weighted standard deviations σ_w reported in Tables 3 and 4 (see Appendices A, B and C for the details of the computation of σ_w). However, the difference is so small (of the order of 10^{-3} m.u. or less) that we may argue whether it represents a clear preference in favour of the purely empirical EXP model with respect to the more physically grounded CBL one.

6 ANALYSIS OF THE M_w – m_b RELATIONSHIP

For the GBL data set, we have 30 326 earthquakes with both m_b and M_w in the range $3.6 \leq m_b \leq 7.2$ (Fig. 7). At low magnitudes ($m_b < 4.5$), the EXP model computed using the CSQ method (red) strongly deviates from the tail of data and tends asymptotically to the line of equality between M_w and m_b . On the contrary, the curve computed by Storchak *et al.* (2013) using the OLS method (GEM GBL, solid blue) well follows such tail and strongly deviates from our EXP model. As evidenced by Gasperini *et al.* (2013b) this difference is likely to be due to both the incompleteness of M_w data in Global catalogues below 5.0–5.5 and the use of the OLS regression method by Storchak *et al.* (2013). In fact, the regression computed using the OLS method on our GBL data set (solid green in Fig. 8) only slightly deviates from the Storchak *et al.* (2013) GEM curve (solid blue). Even in this case a slight increase (within the rounding error) of their b coefficient (GEM mod, dashed blue) can reduce the differences between the two curves for $m_b > 7.0$ from about 0.2 to 0.3 m.u. down to a few hundredths of m.u. On the contrary the differences for $m_b < 4.5$ probably reflect the use of slightly different data sets between us and Storchak *et al.* (2013).

The fitting of the CBL-R model for all m_b data sets showed that the free estimate of Δ systematically hits any reasonable upper limit we can set. Note that it is not possible to extend such limit at wish because otherwise the two transition points exceed the range of magnitude data.

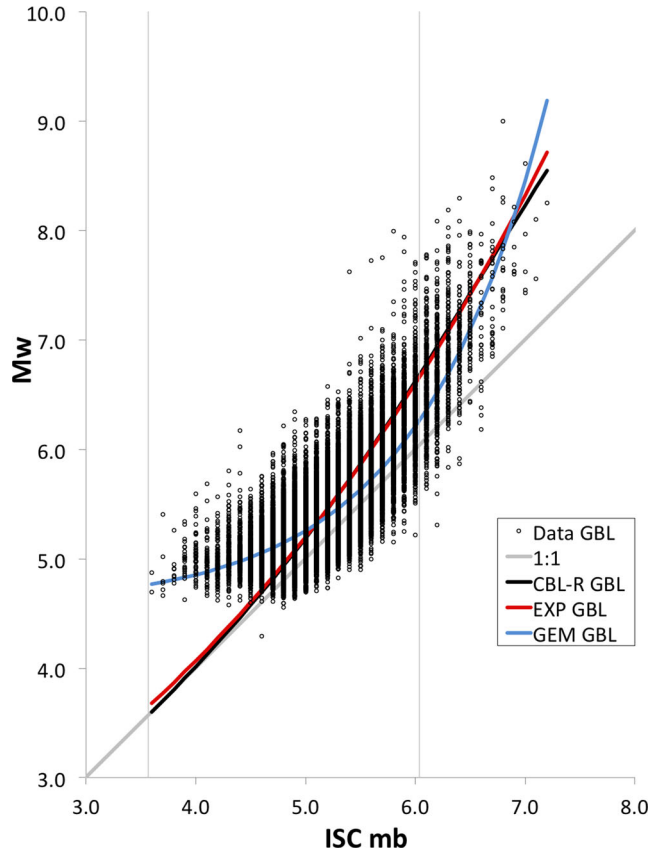


Figure 7. M_w – m_b data distribution and M_w – m_b regression curves for the GBL data set. The two vertical grey lines indicate the transition magnitudes ($m_{b \text{ low}}$ and $m_{b \text{ up}}$) between the linear traits and the connecting arc of the CBL-R model.

We then fixed $\Delta = 2$ (corresponding to a range between $m_{b \text{ up}}$ and $m_{b \text{ low}}$ of about 2.5 m.u.) and estimated only the other two parameters from the data. The resulting CBL-R curve for the GBL area (black in Fig. 7) well coincides with the EXP one for $4.2 < m_b < 6.7$ and deviates at lower and larger magnitudes where, however, both regressions are poorly constrained by data. The slope at high magnitudes (1.609 ± 0.018) is definitely lower than the theoretical value (2) suggested by Kanamori & Anderson (1975). This might indicate that the transition to the theoretical value would occur for m_b larger than those that are usually computed. In any case we can exclude that the m_b data set is incomplete at large magnitudes because the large majority of earthquakes with $M_w > 7.0$ have an m_b estimate in the ISC data set.

For the MED area we have 1673 earthquakes with both m_b and M_w (Fig. 9) in the range $3.5 \leq m_b \leq 6.2$. Note that the number of earthquakes is lower than that (2901) reported by Gasperini *et al.* (2013b) because, as noted above, their MED area is larger than that used here. The EXP model fitted using the CSQ method (solid red) almost coincides with that obtained for the GBL data set (dashed green) but now it lies well within the data owing to the higher completeness of the data set, which is integrated with the data of ETHZ and RCMT regional catalogues at low magnitudes. In this case the CBL-R model (with fixed $\Delta = 2$) well coincides (black) with the EXP model and with that computed for the GBL data set over the entire the range of magnitudes. The slope at high magnitudes (1.509 ± 0.055) is lower but reasonably consistent with that computed for the GBL data set (Table 5).

For the ITA area we have 468 earthquakes with both m_b and M_w (Fig. 10) in the range $3.3 \leq m_b \leq 6.1$. For $m_b > 4$, the EXP (solid red) and CBL-R (solid black) models are very similar to each other and to the EXP model computed for the GBL and MED data sets (the latter is not reported in Fig. 10 because it is almost indistinguishable from the GBL one). Some deviation only occurs for the EXP model at low magnitudes (0.12 m.u. for $m_b = 3.3$). This might represent an evidence of a regional peculiarity of the ITA area with respect to the MED and GBL area and/or of the beginning of the transition from slope 1 to 2/3, even if the small number of low magnitude data suggests caution. The slope of the CBL-R model at high magnitudes (1.496 ± 0.145) is lower than those computed for the MED and GBL, but statistically indistinguishable from both of them (Table 5).

For all of the three areas, the EXP model fits the data better than the CBL as indicated by definitely lower weighted standard deviations σ_w reported in Tables 4 and 5. The differences are larger than for M_s and suggests to preferring the EXP model with respect to the CBL-R one when converting m_b to M_w .

The CBL-R model, however, is useful to make some further inferences on the physical nature of the relationship between M_w and m_b . Although there is a substantial agreement among the three data sets on a value around 1.5–1.6 of the slopes of the CBL-R model at high magnitudes, we can argue that this might be an artefact of the limited range of m_b (about 2.5 m.u.) within which transition of the slope from 1 to 2 is allowed to occur using $\Delta = 2$.

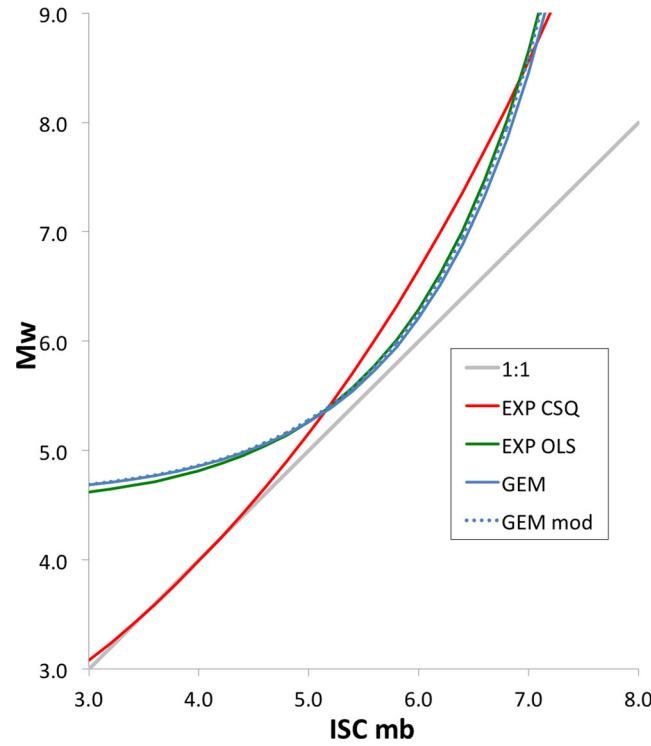


Figure 8. Comparison between different M_w – m_b regression curves for the GBL data set. GEM mod (dashed blue line) indicates the curve computed using a coefficient b slightly modified (0.8649) with respect to the value (0.86) reported by Storchak *et al.* (2013).

We can note that the slope of the EXP models

$$\text{slope} = b \exp(a + bm_b) \quad (9)$$

computed from the GBL and MED coefficients of the CBL model at the maximum observed $m_b = 7.2$ are both very close to the theoretical value of 2 (1.993 and 2.005, respectively). As well, if we extend the range of slope transition to about 3.5 m.u. (by setting $\Delta = 3$) for the GBL model (the same cannot be done for the other two data sets, owing to the limited range of m_b data), the slope of the CBL becomes 2.071 ± 0.055 (Table 5). Hence we might argue that probably the transition from slope 1 to 2 is complete only at the largest m_b 's commonly computed.

7 UNCERTAINTIES OF PROXY MAGNITUDES

The uncertainties of M_w proxies obtained from M_s and m_b can be computed using the formulations reported in Appendices A, B and C. In order to avoid numerical inconsistencies due to rounding errors, we used covariance matrices with seven figures (reported in the online Supporting Information).

In Fig. 11 we show, for both the EXP (top panel) and CBL models, 1σ confidence limits (c.l.) of the regression equations (thin dashed lines) as well as the uncertainties of M_w proxies computed from M_s for the GBL data set, superimposed to the magnitude residuals between observed and proxy M_w . The uncertainty of predicted proxies are computed by A) not including (see discussion in Appendix A) the variance of M_w observations (black thick lines) and B) including such variance (grey). We assumed for all magnitudes the average uncertainty of M_s observations ($\bar{\sigma}_{M_s} = 0.16$) computed over the entire GBL data set (Table 6).

We can note that, owing to the large number of data available, the c.l. of regression equations are almost negligible excepting for the EXP model at large magnitudes. We can also note that the differences between the A (black) and B (grey) approaches are small and become almost negligible at large magnitudes where the variance due to the M_s uncertainty becomes larger owing to the steeper slope of the regression curve.

In Fig. 12 we show the uncertainties of M_w proxies (black) (using the A approach) and the c.l. of regressions (grey) as a function of teleseismic magnitudes for various regression models and data sets. For M_s , the proxy uncertainties are larger for MED and ITA data sets with respect to the GBL one owing to the larger average uncertainty of M_s observations (Table 6) and to the larger c.l. of the regressions. For m_b instead, the uncertainties of the GBL data set are larger than that of MED and ITA data sets reflecting the definitely higher heterogeneity of m_b estimates at the global scale. In general the proxy uncertainties vary in the range from 0.10 to 0.25 m.u. for M_s and from 0.20 to 0.45 for m_b .

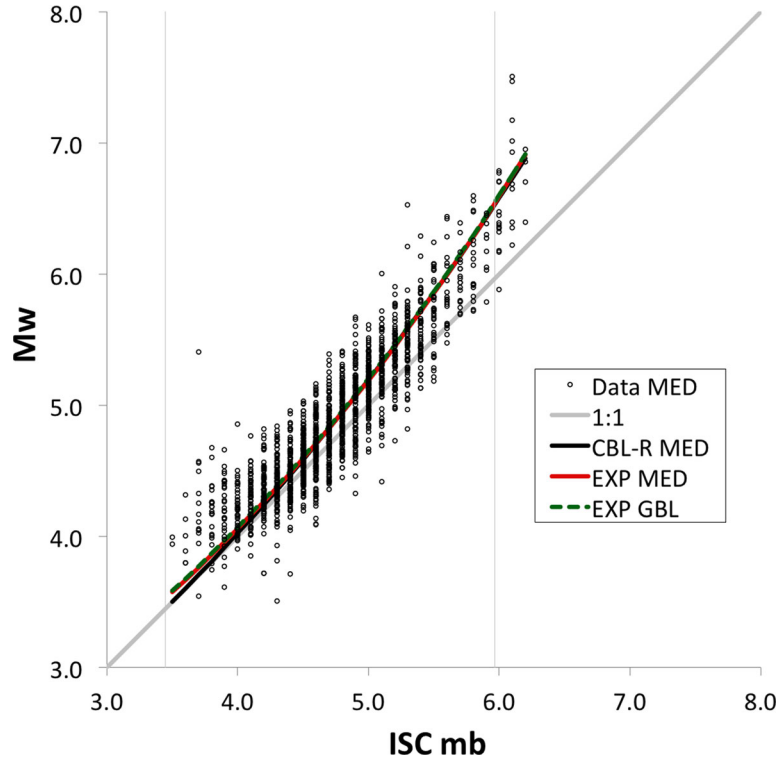


Figure 9. Same as Fig. 7 for the MED data set.

Table 5. CSQ regression parameters, standard deviations (68 per cent) and covariances of the CBL-R model (eq. 4) for m_b , for different data sets. σ_w indicates the weighted standard deviation of regression (see Appendix C).

Data set	a	b	Δ	$m_{b\ c}$	$m_{b\ c}$	r	$m_{b\ low}$	$m_{b\ up}$	σ_w
GBL	1.609 ± 0.018	-3.031 ± 0.104	2	-8.719	15.853	17.375	3.57	6.04	0.3396
MED	1.509 ± 0.055	-2.475 ± 0.306	2	-10.634	17.530	19.915	3.45	5.97	0.2445
ITA	1.534 ± 0.155	-2.684 ± 0.869	2	-10.806	17.939	20.326	3.57	6.09	0.2306
GBL	2.071 ± 0.041	-6.067 ± 0.257	3	-8.981	16.063	17.709	3.54	6.97	0.3391

8 ANALYSIS OF THE FREQUENCY-MAGNITUDE DISTRIBUTION OF THE GLOBAL CATALOGUE HOMOGENIZED IN TERMS OF M_w

We analysed the distribution of M_w proxies of teleseismic magnitudes from the ISC catalogue using the GBL coefficients of (i) the EXP model for both M_s and m_b and of (ii) the CBL model for M_s and the EXP model for m_b .

For comparison, we also analysed the magnitude distribution of the IMW catalogue of really observed M_w 's and of a composite catalogue in which real M_w observations from the IMW data set, when available, are substituted to proxies. In this case, to match the really observed M_w from the IMW data set with the M_w proxies from the ISC data set, we used the same time window (10 s) adopted for building the data sets to regress (see Section 3) and a wider spatial window corresponding the subsurface rupture length of the earthquake estimated from M_w using the empirical relationships by Wells & Coppersmith (1994) (with a minimum of 20 km). This large spatial window reduces the risk of missing the match between the two data sets (particularly for large earthquakes) but implies that in some cases the match found might be wrong.

When both M_s and m_b are available for the same earthquake, we estimated an average M_w proxy weighted with the inverses of the respective variances

$$M_{ave} = \sum \frac{M_i}{\sigma_i^2} / \sum \frac{1}{\sigma_i^2}, \quad (10)$$

where M_i and σ_i^2 are the available M_w proxies and their respective variances. We computed as well the uncertainty of the weighted average from the error propagation law as

$$\sigma_{ave} = \sqrt{1 / \sum \frac{1}{\sigma_i^2}}. \quad (11)$$

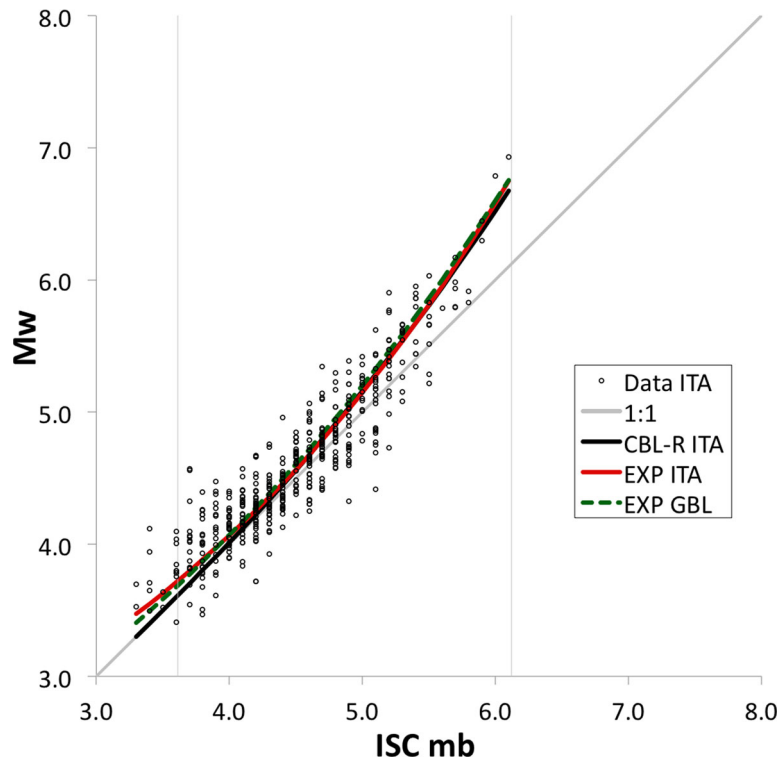


Figure 10. Same as Fig. 7 for the ITA data set.

In the time interval from 1964 to 2010, the ISC data set includes more than 480 000 earthquakes for which an M_w proxy can be computed from M_s , m_b or both.

The completeness of the ISC catalogue and in particular of teleseismic magnitude estimates made by the ISC was poor during the first decades after 1964 but increased significantly in more recent times. Hence, to analyse the frequency–magnitude distribution of the catalogues resulting from the application of our magnitude conversion formulas, we considered a narrower time interval from 1990 to 2010, during which even the IMW data set, used for comparison, became more complete. In this time window we have 337 711 earthquakes for which an M_w proxy can be computed from M_s , m_b or both.

In Fig. 13, we show the cumulative and the differential (binned) frequency–magnitude distribution (Gutenberg & Richter 1944, now on G&R) for the GBL reference data set of really observed M_w from the IMW data set. The cumulative distribution was plotted as the inverse ordering rank (from the largest to the smallest) of each magnitude estimate as a function of M_w while the differential distribution was plotted as the numbers of earthquakes within bins of 0.1 m.u. with respect to the central magnitude of each bin. Both plots were normalized to the length of the time interval in years (21). We also plotted in Fig. 13 the best-fitting lines computed according the maximum likelihood approach (Aki 1965) corrected for the data binning (see for details Gasperini *et al.* 2013a). The vertical dashed line indicates the assumed completeness magnitude threshold ($M_w = 5.3$) for computing the G&R straight lines. In the upper-right inset we display the behaviour of the completeness rate, defined as the ratio between observed and predicted numbers of data with M_w equal to or larger than a given $M_{\min}$. In the lower-left inset we show instead the b -value as a function of the cut-off magnitude $M_{\min}$. We can note the high homogeneity of this data set, showing very stable b -values and completeness rates very close to 100 per cent within the magnitude range $5.3 \leq M_w \leq 7.6$.

In Fig. 14 we show the same plots for the data sets of M_w proxies computed from the EXP model for both M_s and m_b (top panel) and from the CBL model for M_s and the EXP model for m_b (bottom panel). For the assumed completeness threshold ($M_w = 4.8$), both data sets show a b -value definitely larger than that of the IMW data set and even larger than the slope of the curve for magnitudes larger than about 6. This is also reflected by the behaviour of the completeness rate indicating, for $M_{\min} > 6$, a clear overcompleteness for the EXP model and to a lesser amount for the CBL-EXP one. We may argue that this is an effect of the incompleteness of the M_w – M_s GBL data set at low magnitudes discussed above because, if the coefficients of the EXP M_w – M_s relationship computed for the MED data set (which is much more complete at low magnitudes) are used for computing the M_w proxies from $M_s \leq 5.5$ (Fig. 15 top), both the b -value and the completeness rates decrease and their behaviours as a function of $M_{\min}$ become more stable.

Moreover, if observed M_w 's are substituted (when available) to EXP proxies (Fig. 15 bottom), the overestimation of completeness rate at large magnitudes almost totally disappears and its behaviour becomes quite stable (around 95 per cent) within the magnitude range $4.8 \leq M_w \leq 7.6$. We verified instead that a similar procedure applied to the CBL-EXP data set does not improve the stability of the b -value and of the completeness rate as function of $M_{\min}$.

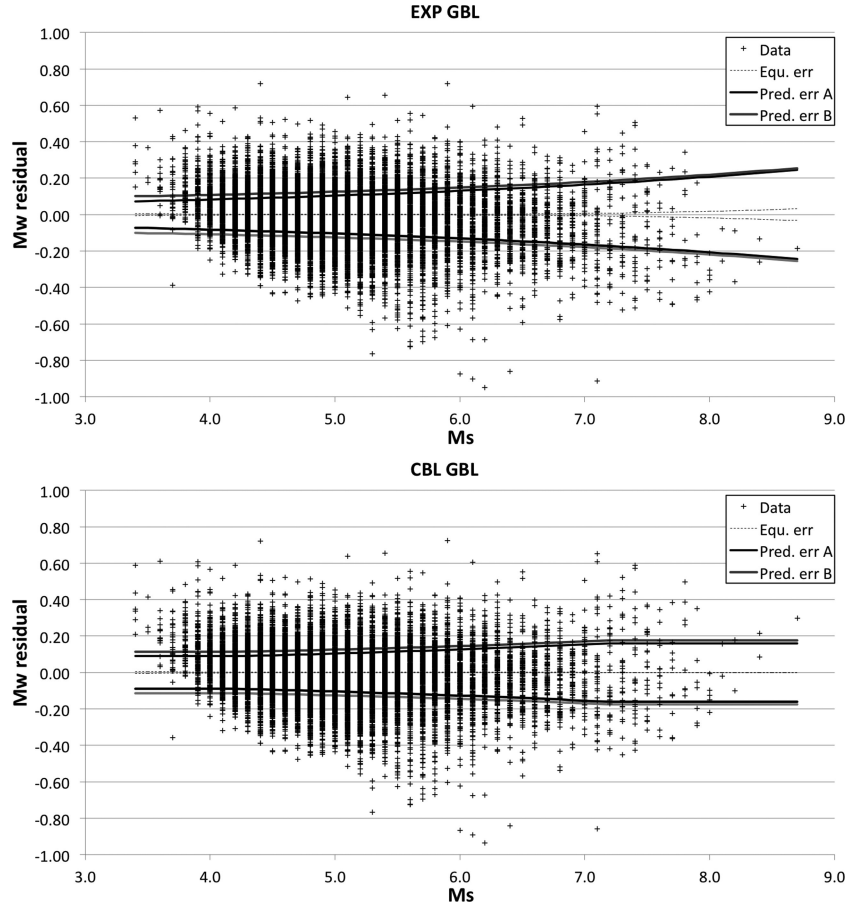


Figure 11. Uncertainties of M_s proxies computed for the GLB data set, using both the EXP (top panel) and CBL models (bottom panel), superimposed to the magnitude residuals between observed and proxy M_w . The thin dashed lines indicate 1σ confidence limits (c.l.) of the regression equation while the thick solid lines the uncertainties of predicted proxies computed (A) not including (see discussion in Appendix A) the variance of M_w observations (black) and (B) including such variance (grey).

Table 6. Averages uncertainties of teleseismic magnitude estimates used for plotting confidence limits for various data sets.

Data set	GBL	MED	ITA
$\bar{\sigma}_{M_s}$	0.16	0.19	0.19
$\bar{\sigma}_{m_b}$	0.24	0.19	0.18

9 CONCLUSIONS

We analysed the problem of converting teleseismic magnitudes M_s and m_b , provided by the online Bulletin of the ISC, to moment magnitude M_w using curvilinear regression laws and the CSQ method (Stromeyer *et al.* 2004) that properly account for the non-linearity of the relationships and for the uncertainties of regressed magnitudes. We used both an exponential model (EXP) recently proposed by Storchak *et al.* (2013) and two circularly connected bilinear models (CBL for M_s and CBL-R for m_b) that are analogous to the model proposed by Ekström & Dziewonski (1988) for M_s . For all of the regression models, we also derived the relevant formulations to compute the uncertainties of M_w proxies.

We obtained the initial uncertainties of teleseismic magnitudes, required for the correct application of the CSQ orthogonal regression method, from the analysis of the statistical dispersion of magnitude estimates from different stations and of the spatial variations of average M_s and m_b residuals. We showed that the standard deviation of the mean magnitude alone underestimates the real uncertainty and that an additional term related to the spatial variability of wave propagation properties in the crust and the lithosphere is also required to reliably describe such uncertainty. In fact, the average residuals of M_w proxies for both M_s and m_b show a clear correlation with the major geodynamic features of the Earth: positive (M_w proxies overestimate measured M_w) under the continents and negative (M_w proxies underestimate measured M_w) under the oceans. This suggests a possible future improvement of the conversion between teleseismic magnitudes and M_w using regionalized residuals.

For M_s , we found significant differences between the regression curves obtained at a global scale (GBL), using a data set of M_w estimates provided by the Global CMT catalogue (GCMT) and by the MT catalogue of the National Earthquake Information Centre (NEIC), and the

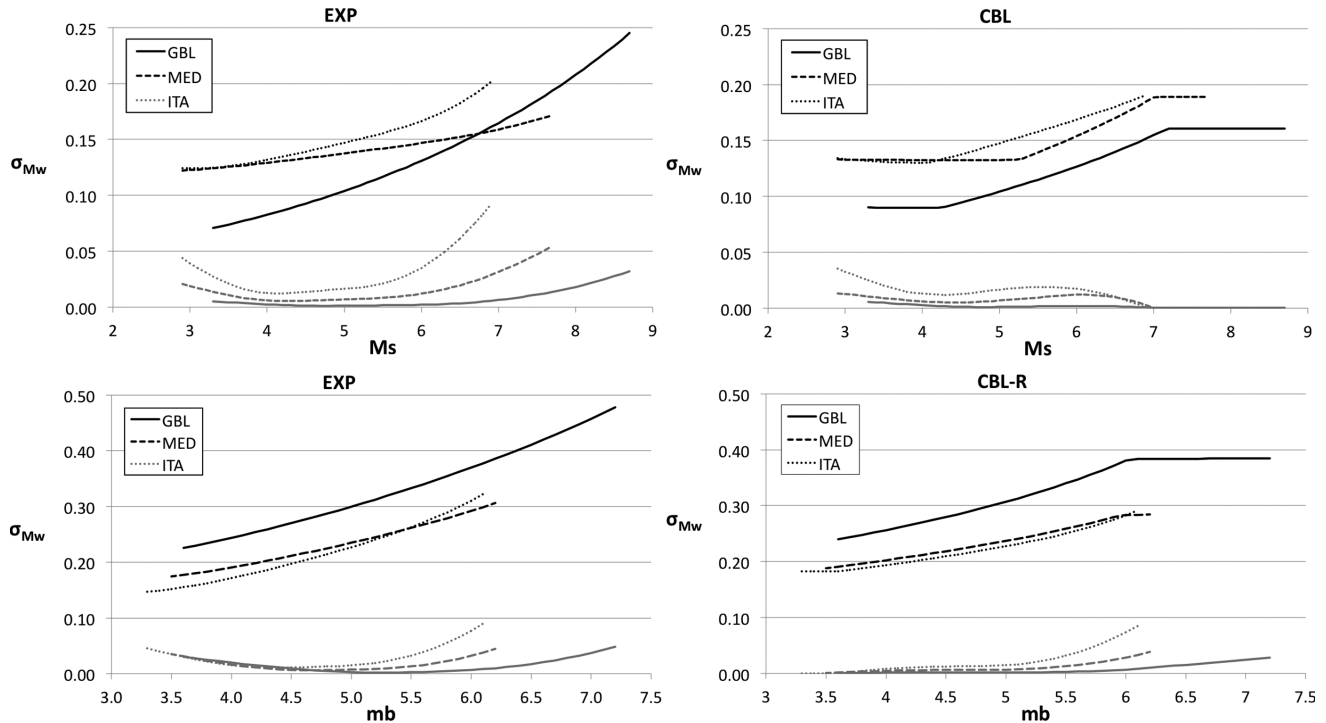


Figure 12. Uncertainties of M_w proxies (black) and c.l. of regressions (grey) as a function of teleseismic magnitude for various models and data sets.

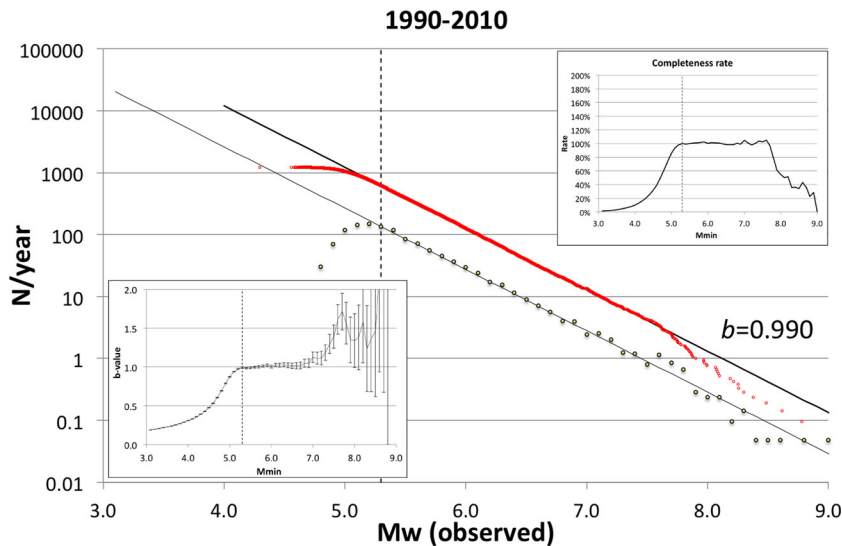


Figure 13. Cumulative (red dots) and differential (black circles) frequency–magnitude distribution of the GBL catalogues from 1990 to 2010 of real M_w estimates from the IMW data set. Plots are normalized to the length of the time interval in years (21). In the upper-right inset the behaviour of the ratio between observed and predicted numbers of data with M_w proxy equal to or larger than a given $M_{\min}$ (completeness rate). In the lower-left insets the b -value computed as a function of the assumed cut-off magnitude ($M_{\min}$). The vertical dashed lines indicate the assumed completeness threshold (5.3).

regression curves obtained at the Euro-Mediterranean scale (MED), using an integrated M_w data set, which completeness for $M_w < 5.0$ – 5.5 is improved by the addition of M_w estimates provided by the MT catalogue of the Eidgenössische Technische Hochschule Zürich (ETHZ) and by the European-Mediterranean Regional Centroid Moment Tensor catalogue (RCMT) of the Istituto Nazionale di Geofisica e Vulcanologia (INGV). Similar differences were also found with respect to regressions made using an Italian data set (ITA) also including the M_w estimates provided by the Time Domain Moment Tensor catalogue of the INGV.

At low magnitudes ($M_w < 5.0$ – 5.5), the slopes of the regression curves computed from the less complete GBL data set are lower than those obtained from the MED and ITA data sets and this produces an overestimation of M_w proxies from M_s up to about 0.5 m.u. (for $M_s = 3.5$) computed from GBL coefficients.

At high magnitudes ($M_w > 7.0$), while the EXP curve computed from the GBL data set lies well above the line of equality between M_w and M_s , the corresponding MED curve lies slightly below the equality line. The latter peculiar feature of the MED data set is quite surprising because all evidences would indicate that at large magnitudes M_s saturates and substantially underestimates M_w . We could argue, however,

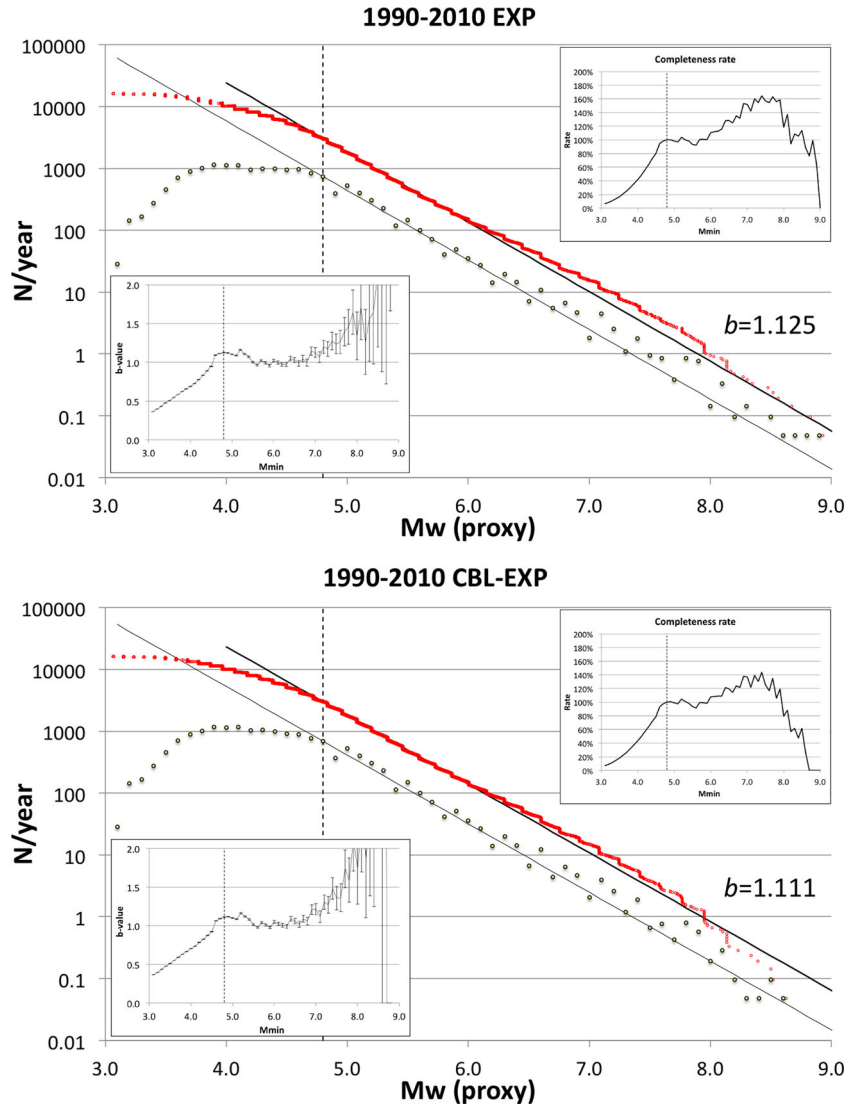


Figure 14. Same as Fig. 13 for the GBL catalogue of M_w proxies computed using the EXP models (top panel) for both M_s and m_b and using the CBL model for M_s and the EXP one for m_b (bottom panel). The vertical dashed lines indicate the assumed completeness threshold (4.8).

that this might be a spurious effect related to the low number (6) of earthquakes with $M_w > 7.0$ in the MED data set. In this case, the CBL curve, which coincides with the line of equality between M_w and M_s , might better represent the physical reality in the MED area at high magnitudes. Conversely, the CBL curve for the GBL data set does not account well for the saturation of M_s at large magnitudes. In general, the results of the CBL models for the regression between M_w and M_s substantially confirm that real data approximately follow the theoretical slopes of 2/3 and 1 at low and high magnitudes respectively and that a significant deviation from the 1:1 curve occurs for the GBL data set only at $M_s > 7.5$ –8.0.

For m_b , the differences between the regression computed for the GBL and MED data sets are almost negligible for both the EXP and CBL-R models. A small difference of the ITA data set with respect to both GBL and MED ones can be observed for the EXP model at very low magnitudes and consists of a slight overestimation of M_w with respect to m_b (0.12 m.u. at most). We might argue that this represents a weak evidence of the transition to slope 2/3, but the scarce number of data does not allow to draw a final conclusion. In general, our results indicate that the transition between theoretical slopes 1 and 2 occurs over a wide range of magnitudes and probably completes only at the largest m_b 's (7.2) included in our data set.

We analysed the frequency–magnitude distribution (Gutenberg & Richter 1944) of the M_w average proxies computed for the GBL data set using the regression coefficients for both the EXP and CBL models. We found that the b -value of the catalogue of proxies resulting from the simple application of the EXP coefficients computed for the GBL area is rather variable for cut-off magnitudes ranging from 4.8 to 6.0 and is generally larger than that of the integrated data set of directly measured M_w (IMW). This also holds but to a minor extent for the M_w proxies computed using the GBL coefficients of the CBL model for M_s . A more stable behaviour of the b -value can be reached by using, even for the GBL data set, the coefficients of the EXP model derived for the MED area for converting $M_s \leq 5.5$ to M_w . In summary, our final recommendations for computing M_w proxies are the following:

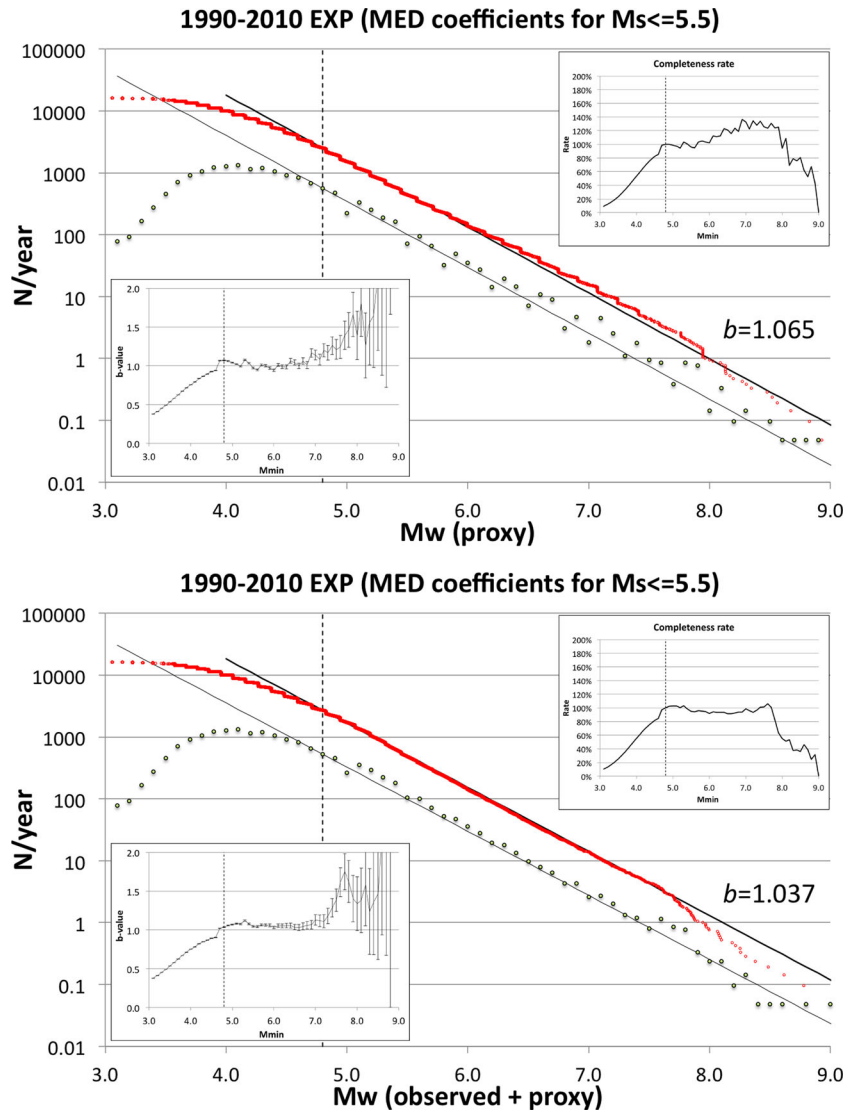


Figure 15. Same as Fig. 13 for the catalogue of M_w proxies (top panel) and of the combination of observed and proxy M_w 's (bottom panel) computed using MED coefficients for $M_s \leq 5.5$.

(1) at the global scale

$$\begin{cases} M_w = \exp(2.133 + 0.063M_s) - 6.205, & M_s \leq 5.5 \\ M_w = \exp(-0.109 + 0.229M_s) + 2.586, & M_s > 5.5 \end{cases} \quad (12)$$

$$M_w = \exp(0.741 + 0.210m_b) - 0.785 \quad (13)$$

(2) at the Euro-Mediterranean scale

$$M_w = \exp(2.133 + 0.063M_s) - 6.205 \quad (14)$$

$$M_w = \exp(0.719 + 0.212m_b) - 0.737. \quad (15)$$

(3) at the Italian scale

$$M_w = \exp(1.421 + 0.108M_s) - 1.863 \quad (16)$$

$$M_w = \exp(0.058 + 0.284m_b) - 0.770. \quad (17)$$

Owing to the almost negligible differences found, the global m_b eq. (13) can be confidently used even at the MED and ITA scales. If we consider unphysical the ‘undersaturation’ observed at the MED scale for large M_s , a valid alternative is the use of CBL coefficients (reported in Table 4) in place of the EXP ones.

ACKNOWLEDGEMENTS

We thank three anonymous referees for thoughtful comments and suggestions. One of the authors (BL) has benefited from funding provided by the Italian Presidenza del Consiglio dei Ministri – Dipartimento della Protezione Civile (DPC). Scientific papers funded by DPC do not represent its official opinion and policies.

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APPENDIX A: CHI-SQUARE REGRESSION FOR THE EXPONENTIAL (EXP) REGRESSION MODEL

According to Gasperini *et al.* (2013b), the chi-square statistics to be minimized for the exponential model of eq. (2) is

$$\chi^2 = \sum_{i=1}^N \frac{[M_{wi} - \exp(a + bx_i) - c]^2}{b^2 \exp(2a + 2bx_i) \sigma^2(x_i) + \sigma^2(M_{wi})}, \quad (\text{A1})$$

where M_{wi} and x_i are observations of M_w and of teleseismic magnitudes (M_s and m_b), respectively and $\sigma(M_{wi})$ and $\sigma(x_i)$ their respective uncertainties.

Gasperini *et al.* (2013b) showed how the value of c minimizing χ^2 can be determined analytically so that only a and b need to be computed numerically. An estimate of the goodness-of-fit of the regression can be computed as the overall unbiased weighted variance of the regression

$$\sigma_w^2 = \frac{N}{N-3} \sum_{i=1}^N \frac{[M_{wi} - \exp(a + bx_i) - c]^2}{b^2 \exp(2a + 2bx_i) \sigma^2(x_i) + \sigma^2(M_{wi})} / \sum_{i=1}^N \frac{1}{b^2 \exp(2a + 2bx_i) \sigma^2(x_i) + \sigma^2(M_{wi})}. \quad (\text{A2})$$

The variance covariance matrix of regression coefficients, normalized to the empirical regression standard deviation (see Lolli & Gasperini 2012), can be computed by inverting the finite-difference Hessian of χ^2 at the maximum

$$\mathbf{V}(a, b, c) = 2 \frac{\chi^2}{N-3} \begin{vmatrix} \frac{\partial \chi^2}{\partial a \partial a} & \frac{\partial \chi^2}{\partial a \partial b} & \frac{\partial \chi^2}{\partial a \partial c} \\ \frac{\partial \chi^2}{\partial b \partial a} & \frac{\partial \chi^2}{\partial b \partial b} & \frac{\partial \chi^2}{\partial b \partial c} \\ \frac{\partial \chi^2}{\partial c \partial a} & \frac{\partial \chi^2}{\partial c \partial b} & \frac{\partial \chi^2}{\partial c \partial c} \end{vmatrix}^{-1} = \begin{vmatrix} \sigma_a^2 & \text{cov}(a, b) & \text{cov}(a, c) \\ \text{cov}(a, b) & \sigma_b^2 & \text{cov}(b, c) \\ \text{cov}(a, c) & \text{cov}(b, c) & \sigma_c^2 \end{vmatrix} \quad (\text{A3})$$

From which the uncertainties of the regression coefficients can be computed as the square roots of the corresponding diagonal elements.

According to error propagation law (see Gasperini *et al.* 2013a), the (standard) uncertainties σ_{M_w} of magnitude proxies computed using eq. (2) can be determined as

$$\begin{aligned} \sigma_{M_w}^2 &= \frac{\partial M_w(x)}{\partial a} \frac{\partial M_w(x)}{\partial a} \sigma_a^2 + \frac{\partial M_w(x)}{\partial b} \frac{\partial M_w(x)}{\partial b} \sigma_b^2 + \frac{\partial M_w(x)}{\partial c} \frac{\partial M_w(x)}{\partial c} \sigma_c^2 + 2 \frac{\partial M_w(x)}{\partial a} \frac{\partial M_w(x)}{\partial b} \text{cov}(a, b) \\ &+ 2 \frac{\partial M_w(x)}{\partial a} \frac{\partial M_w(x)}{\partial c} \text{cov}(a, c) + 2 \frac{\partial M_w(x)}{\partial b} \frac{\partial M_w(x)}{\partial c} \text{cov}(b, c) + \frac{\partial M_w(x)}{\partial x} \frac{\partial M_w(x)}{\partial x} \sigma^2(x). \end{aligned} \quad (\text{A4})$$

By substituting the partial derivatives, eq. (A4) can be rewritten as

$$\sigma_{M_w}^2 = \exp(2a + 2bx) [\sigma_a^2 + x^2 \sigma_b^2 + b^2 \sigma^2(x) + 2x \text{cov}(a, b)] + 2 \exp(a + bx) [\text{cov}(a, c) + x \text{cov}(b, c)] + \sigma_c^2. \quad (\text{A5})$$

A Fortran subroutine for computing the M_w proxy and the associated uncertainty for the EXP model is available online from the Supporting Information.

Note that eqs (A4) and (A5), which we derived from the error propagation law, include the terms accounting for the variance due to regression parameters as well as the term accounting for the variance of the independent variable $\sigma^2(x)$ but do not include instead the variance of the dependent variable $\sigma^2(M_w)$.

This differs from the usual treatment of OLS regressions (i.e. Draper & Smith 1981) where the variance of the independent variable is by definition 0 and the prediction variance is obtained by summing the variance of the dependent variable (assumed to coincide with the empirical variance of regression residuals) to the variance due to the regression equation. Such approach is strictly valid only if the variance of the independent variable is really 0 (or negligible) and is reasonably conservative when the OLS regression is applied despite the hypothesis of null errors in the independent variable is not fully verified (as frequently occurs in scientific practice).

We discussed this problem in Appendix B of Gasperini *et al.* (2013a) where we also considered the possibility to add the variance of the dependent variable $\sigma^2(M_w)$ to $\sigma^2_{M_w}$ that substantially corresponds to the Draper & Smith (1981) approach. However we argued that the scope of magnitude conversions is not to predict the result of a hypothetical experiment (e.g. computing M_w from moment tensor inversion) which would have been observed if the required data (e.g. broad-band waveforms) were available for a given earthquake but rather to estimate the seismic moment M_0 of the earthquake (M_w is by definition a simple logarithmic transformation of M_0), provided that a given type of magnitude (e.g. M_L , M_s , m_b) has been measured. Then we concluded that the variance of the dependent variable has not to be included in the variance of the predicted proxy magnitudes.

In any case, in this work we considered both approaches: (A) not including the variance of the dependent variable as in eqs (A4) and (A5) and (B) including it by summing $\sigma^2(M_w)$ to such equations. As the uncertainty of M_w is small (0.07–0.10 m.u.) the difference between the two approaches is also small.

APPENDIX B: CHI-SQUARE REGRESSION FOR THE CIRCULARLY CONNECTED BILINEAR (CBL) REGRESSION MODEL (FOR M_s).

The two straight lines of eq. (3) intersect at (see Fig. 1)

$$M_{sI} = M_{wI} = \frac{b}{1-a}. \quad (B1)$$

The coordinates in the M_s – M_w plane of two points located along the two lines at the same distance Δ from the intersection point are

$$\begin{aligned} M_{s \text{ low}} &= M_{sI} - \frac{\Delta}{\sqrt{1+a^2}} \\ M_{w \text{ low}} &= a M_{s \text{ low}} + b \\ M_{s \text{ up}} &= M_{sI} + \frac{\Delta}{\sqrt{2}} \\ M_{w \text{ up}} &= M_{s \text{ up}}. \end{aligned} \quad (B2)$$

The equations of two straight lines perpendicular to the two previous lines and passing through the two points are

$$\begin{aligned} M_w &= -\frac{1}{a} M_s + a M_{s \text{ low}} + b + \frac{1}{a} M_{s \text{ low}} \\ M_w &= -M_s + 2 M_{s \text{ up}} \end{aligned} \quad (B3)$$

they intersect at

$$\begin{aligned} M_{sc} &= \frac{(1+a^2)M_{s \text{ low}} + ab - 2a M_{s \text{ up}}}{1-a} \\ M_{wc} &= -M_{sc} + 2 M_{s \text{ up}}. \end{aligned} \quad (B4)$$

Such intersection point is taken as the centre of the circle connecting the two regression lines, which radius can be computed as

$$R = \sqrt{(M_{s \text{ low}} - M_{sc})^2 + (M_{w \text{ low}} - M_{wc})^2}. \quad (B5)$$

It is easy to demonstrate that the equation of such circle (taken with upward concavity, owing to the geometry of the problem)

$$M_w(M_s) = M_{wc} - \sqrt{R^2 - (M_s - M_{sc})^2} \quad (B6)$$

coincides with the first straight line in $(M_{s \text{ low}}, M_{w \text{ low}})$ and with the second line in $(M_{s \text{ up}}, M_{w \text{ up}})$ as well as that its first derivative coincides with the slope of the first straight line in $(M_{s \text{ low}}, M_{w \text{ low}})$ and with the slope of the second line in $(M_{s \text{ up}}, M_{w \text{ up}})$.

The non-linear regression problem consists of finding the coefficients a and b and the distance Δ that fit the data at best over the entire range of magnitudes. According to Stromeyer *et al.* (2004), the chi-square statistics to be minimized is

$$\chi^2 = \sum_{i=1}^N \frac{[M_{wi} - M_w(M_{si})]^2}{\sigma^2[M_{wi} - M_w(M_{si})]}, \quad (B7)$$

where $\sigma^2[M_{wi} - M_w(M_{si})]$ is the expected (*a priori*) variance of the regression residuals. Under the assumption that the variances of the two variables are uncorrelated, $\sigma^2[M_{wi} - M_w(M_{si})]$ of a generic observation can be computed, as a function of the *a priori* uncertainties $\sigma(M_s)$

and $\sigma(M_w)$ of the two magnitude observations, by the error propagation law, as

$$\begin{aligned}\sigma^2[M_{wi} - M_w(M_{si})] &\approx \left[\frac{\partial [-M_w(M_s)]}{\partial M_s} \sigma(M_{si}) \right]^2 + \left[\frac{\partial [-M_w(M_s)]}{\partial M_w} \sigma(M_{wi}) \right]^2 \\ &= [M'_w(M_{si})]^2 \sigma^2(M_{si}) + \sigma^2(M_{wi}),\end{aligned}\quad (\text{B8})$$

where $M_w(M_s)$ is given by eq. (3) and $M'_w(M_s)$ is

$$M'_w(M_s) = \begin{cases} a, & M_s < M_{s\text{low}} \\ \frac{M_s - M_{sc}}{\sqrt{R^2 - (M_s - M_{sc})^2}}, & M_{s\text{low}} \leq M_s \leq M_{s\text{up}} \\ 1, & M_s > M_{s\text{up}} \end{cases} \quad (\text{B9})$$

Hence, eq. (B7) can be rewritten as

$$\chi^2 = \sum_{i=1}^N \frac{[M_{wi} - M_w(M_{si})]^2}{[M'_w(M_{si})]^2 \sigma^2(M_{si}) + \sigma^2(M_{wi})} \quad (\text{B10})$$

from which the best-fitting regression parameters a , b and Δ can be computed by numerical minimization. An estimate of the goodness-of-fit of the regression can be computed as the unbiased overall weighted variance of the regression

$$\sigma_w^2 = \frac{N}{N-3} \sum_{i=1}^N \frac{[M_{wi} - M_w(M_{si})]^2}{[M'_w(M_{si})]^2 \sigma^2(M_{si}) + \sigma^2(M_{wi})} \bigg/ \sum_{i=1}^N \frac{1}{[M'_w(M_{si})]^2 \sigma^2(M_{si}) + \sigma^2(M_{wi})}. \quad (\text{B11})$$

The variance covariance matrix of regression coefficients, normalized to the regression standard deviation, can be computed (see Lolli & Gasperini 2012) by inverting the numerical Hessian of χ^2 at the maximum as

$$\mathbf{V}(a, b) = 2 \frac{\chi^2}{N-3} \begin{vmatrix} \frac{\partial \chi^2}{\partial a \partial a} & \frac{\partial \chi^2}{\partial a \partial b} & \frac{\partial \chi^2}{\partial a \partial \Delta} \\ \frac{\partial \chi^2}{\partial b \partial a} & \frac{\partial \chi^2}{\partial b \partial b} & \frac{\partial \chi^2}{\partial b \partial \Delta} \\ \frac{\partial \chi^2}{\partial \Delta \partial a} & \frac{\partial \chi^2}{\partial \Delta \partial b} & \frac{\partial \chi^2}{\partial \Delta \partial \Delta} \end{vmatrix}^{-1} = \begin{vmatrix} \sigma_a^2 & \text{cov}(a, b) & \text{cov}(a, \Delta) \\ \text{cov}(a, b) & \sigma_b^2 & \text{cov}(b, \Delta) \\ \text{cov}(a, \Delta) & \text{cov}(b, \Delta) & \sigma_\Delta^2 \end{vmatrix}. \quad (\text{B12})$$

From which the uncertainties σ_a , σ_b and σ_Δ of the regression parameters can be computed as square roots of the corresponding diagonal elements.

According to error propagation law, the (standard) uncertainties of M_w proxies computed using eq. (3) can be estimated as

$$\begin{aligned}\sigma_{M_w}^2 &= \frac{\partial M_w(M_s)}{\partial a} \frac{\partial M_w(M_s)}{\partial a} \sigma_a^2 + \frac{\partial M_w(M_s)}{\partial b} \frac{\partial M_w(M_s)}{\partial b} \sigma_b^2 + \frac{\partial M_w(M_s)}{\partial \Delta} \frac{\partial M_w(M_s)}{\partial \Delta} \sigma_\Delta^2 + 2 \frac{\partial M_w(M_s)}{\partial a} \frac{\partial M_w(M_s)}{\partial b} \text{cov}(a, b) \\ &+ 2 \frac{\partial M_w(M_s)}{\partial a} \frac{\partial M_w(M_s)}{\partial \Delta} \text{cov}(a, \Delta) + 2 \frac{\partial M_w(M_s)}{\partial b} \frac{\partial M_w(M_s)}{\partial \Delta} \text{cov}(b, \Delta) + \frac{\partial M_w(M_s)}{\partial M_s} \frac{\partial M_w(M_s)}{\partial M_s} \sigma^2(M_s).\end{aligned}\quad (\text{B13})$$

For data along the two linear traits, the partial derivatives in eq. (B13) are relatively simple hence the variances of M_w proxies are

$$\sigma_{M_w}^2 = \begin{cases} M_s^2 \sigma_a^2 + \sigma_b^2 + 2M_s \text{cov}(a, b) + a^2 \sigma^2(M_s), & M_s < M_{s\text{low}} \\ \sigma^2(M_s), & M_s > M_{s\text{up}} \end{cases} \quad (\text{B14})$$

For $M_{s\text{low}} \leq M_s \leq M_{s\text{up}}$, the partial derivatives can be computed analytically as follows:

$$\frac{\partial M_w(M_s)}{\partial M_s} = \frac{(M_s - M_{sc})}{\sqrt{R^2 - (M_s - M_{sc})^2}} \quad (\text{B15})$$

$$\frac{\partial M_w(M_s)}{\partial a} = \frac{\partial M_{wc}}{\partial a} - \frac{R \frac{\partial R}{\partial a} + (M_s - M_{sc}) \frac{\partial M_{sc}}{\partial a}}{\sqrt{R^2 - (M_s - M_{sc})^2}}, \quad (\text{B16})$$

where

$$\frac{\partial M_{sc}}{\partial a} = \frac{2aM_{s\text{low}} + (1+a^2) \frac{\partial M_{s\text{low}}}{\partial a} + b - 2M_{s\text{up}} - 2a \frac{\partial M_{s\text{up}}}{\partial a}}{1-a} + \frac{(1+a^2)M_{s\text{low}} + ab - 2aM_{s\text{up}}}{(1-a)^2}. \quad (\text{B17})$$

$$\frac{\partial R}{\partial a} = \frac{(M_{s \text{ low}} - M_{s \text{ c}}) \left(\frac{\partial M_{s \text{ low}}}{\partial a} - \frac{\partial M_{s \text{ c}}}{\partial a} \right) + (M_{w \text{ low}} - M_{w \text{ c}}) \left(\frac{\partial M_{w \text{ low}}}{\partial a} - \frac{\partial M_{w \text{ c}}}{\partial a} \right)}{R}. \quad (\text{B18})$$

$$\frac{\partial M_{s \text{ low}}}{\partial a} = \frac{b}{(1-a)^2} + \frac{a\Delta}{(1+a^2)\sqrt{1+a^2}} \quad (\text{B19})$$

$$\frac{\partial M_{s \text{ up}}}{\partial a} = \frac{b}{(1-a)^2} \quad (\text{B20})$$

$$\frac{\partial M_{w \text{ c}}}{\partial a} = -\frac{\partial M_{s \text{ c}}}{\partial a} + 2\frac{\partial M_{s \text{ up}}}{\partial a} \quad (\text{B21})$$

$$\frac{\partial M_{w \text{ low}}}{\partial a} = M_{s \text{ low}} + a\frac{\partial M_{s \text{ low}}}{\partial a} \quad (\text{B22})$$

$$\frac{\partial M_w(M_s)}{\partial b} = \frac{1}{1-a} \left[1 - \frac{(M_s - M_{s \text{ c}})}{\sqrt{R^2 - (M_s - M_{s \text{ c}})^2}} \right] \quad (\text{B23})$$

and

$$\frac{\partial M_w(M_s)}{\partial \Delta} = \frac{\partial M_{w \text{ c}}}{\partial \Delta} - \frac{R \frac{\partial R}{\partial \Delta} + (M_s - M_{s \text{ c}}) \frac{\partial M_{s \text{ c}}}{\partial \Delta}}{\sqrt{R^2 - (M_s - M_{s \text{ c}})^2}}, \quad (\text{B24})$$

where

$$\frac{\partial M_{s \text{ c}}}{\partial \Delta} = \frac{1}{1-a} \left[-\sqrt{1+a^2} - \sqrt{2}a \right] \quad (\text{B25})$$

$$\frac{\partial R}{\partial \Delta} = \frac{(M_{s \text{ low}} - M_{s \text{ c}}) \left(\frac{\partial M_{s \text{ low}}}{\partial \Delta} - \frac{\partial M_{s \text{ c}}}{\partial \Delta} \right) + (M_{w \text{ low}} - M_{w \text{ c}}) \left(\frac{\partial M_{w \text{ low}}}{\partial \Delta} - \frac{\partial M_{w \text{ c}}}{\partial \Delta} \right)}{R} \quad (\text{B26})$$

$$\frac{\partial M_{s \text{ low}}}{\partial \Delta} = -\frac{1}{\sqrt{1+a^2}} \quad (\text{B27})$$

$$\frac{\partial M_{s \text{ up}}}{\partial \Delta} = \frac{1}{\sqrt{2}} \quad (\text{B28})$$

$$\frac{\partial M_{w \text{ c}}}{\partial \Delta} = -\frac{\partial M_{s \text{ c}}}{\partial \Delta} + 2\frac{\partial M_{s \text{ up}}}{\partial \Delta} \quad (\text{B29})$$

$$\frac{\partial M_{w \text{ low}}}{\partial \Delta} = -\frac{a}{\sqrt{1+a^2}}. \quad (\text{B30})$$

The uncertainties of M_w proxies within the interval $M_{s \text{ low}} \leq M_s \leq M_{s \text{ up}}$ can then be computed by substituting eqs (B15) to (B30) in eq. (B13). The derivatives have been tested numerically by a comparison with finite difference estimates over the entire range of magnitudes. A Fortran subroutine for computing the M_w proxy and the associated uncertainty for the CBL model is available online from the Supporting Information. As for the EXP model we also considered the case (B) in which the variance $\sigma^2(M_w)$ of the dependent variable is added to eq. (B13).

APPENDIX C: CHI-SQUARE REGRESSION FOR THE REVERSED CIRCULARLY CONNECTED BILINEAR (CBL-R) REGRESSION MODEL (FOR m_b)

The two straight lines of eq. (4) intersect at

$$m_{bI} = M_{wI} = \frac{b}{1-a}. \quad (\text{C1})$$

The coordinates in the m_b - M_w plane of two points, along the two lines, located at the same distance Δ from the intersection point are

$$\begin{aligned} m_{b \text{ up}} &= m_{bI} + \frac{\Delta}{\sqrt{1+a^2}} \\ M_{w \text{ up}} &= am_{b \text{ up}} + b \\ m_{b \text{ low}} &= m_{bI} - \frac{\Delta}{\sqrt{2}} \\ M_{w \text{ low}} &= m_{b \text{ low}}. \end{aligned} \quad (C2)$$

The equations of two straight lines perpendicular to the two previous lines and passing through the two points are

$$\begin{aligned} M_w &= -\frac{1}{a}m_b + am_{b \text{ up}} + b + \frac{1}{a}m_{b \text{ up}} \\ M_w &= -m_b + 2m_{b \text{ low}} \end{aligned} \quad (C3)$$

their intersection is at

$$\begin{aligned} m_{bc} &= \frac{(1+a^2)m_{b \text{ up}} + ab - 2am_{b \text{ low}}}{1-a} \\ M_{wc} &= -m_{bc} + 2m_{b \text{ low}}. \end{aligned} \quad (C4)$$

This point is the centre of the circle connecting the two regression lines with radius is

$$R = \sqrt{(m_{b \text{ up}} - m_{bc})^2 + (M_{w \text{ up}} - M_{wc})^2}. \quad (C5)$$

Note the symmetry of such formulation with respect to that one reported in Appendix B for the CBL model. In the practice they can be derived one to the other by simply assuming a negative Δ and consequently swapping the definitions of upper and lower transition magnitudes.

The chi-square statistics to be minimized (see the analogous derivation in Appendix B) to estimate the regression parameters a , b and Δ is

$$\chi^2 = \sum_{i=1}^N \frac{[M_{wi} - M_w(m_{bi})]^2}{[M'_w(m_{bi})]^2 \sigma^2(m_{bi}) + \sigma^2(M_{wi})}, \quad (C6)$$

where

$$M'_w(m_b) = \begin{cases} 1, & m_b < m_{b \text{ low}} \\ \frac{m_b - m_{bc}}{\sqrt{R^2 - (m_b - m_{bc})^2}}, & m_{b \text{ low}} \leq m_b \leq m_{b \text{ up}} \\ c, & m_b > m_{b \text{ up}} \end{cases} \quad (C7)$$

To compare the goodness-of-fit of the regression with other models we define the overall weighted variance of the regression as

$$\sigma_w^2 = \frac{N}{N-3} \sum_{i=1}^N \frac{[M_{wi} - M_w(m_{bi})]^2}{[M'_w(m_{bi})]^2 \sigma^2(m_{bi}) + \sigma^2(M_{wi})} \bigg/ \sum_{i=1}^N \frac{1}{[M'_w(m_{bi})]^2 \sigma^2(m_{bi}) + \sigma^2(M_{wi})}. \quad (C8)$$

Analogously to the CBL model, the variance covariance matrix of regression coefficients, normalized to the regression standard deviation, can be computed by inverting the numerical Hessian of χ^2 at the maximum while the uncertainties of M_w proxies computed by eq. (4) can be determined as well by the error propagation law. Owing to the symmetry of the two models the relevant formulas can be derived (substituting m_b with M_s) from those reported in Appendix B for the CBL model by simply setting a negative Δ and then swapping $m_{b \text{ low}}$ and $m_{b \text{ up}}$ among each other. A Fortran subroutine for computing M_w proxies and associated uncertainties for the CBL-R model is provided in the Supporting Information.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

(<http://gji.oxfordjournals.org/lookup/suppl/doi:10.1093/gji/ggu264/-/DC1>).

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